CH3_Examples of Distributed Systems

1. **Resources sharing:** The Cambridge Distributed Computer System
2. **Machines automation and control underground monitoring/ control coal mines**
3. **Sensors Systems (Sensors Networks SN, Wireless Sensor Networks WSN)**
4. E banking: Bank of America
5. Internet/World-Wide Web WWW
6. Local Area Network and Intranet
7. Database and Database Management System
8. Automatic Teller Machine Network
9. Mobile Computing
10. Video on demand
11. Traffic controls

1. Distributed Processing Applications Overview

Resource Sharing:

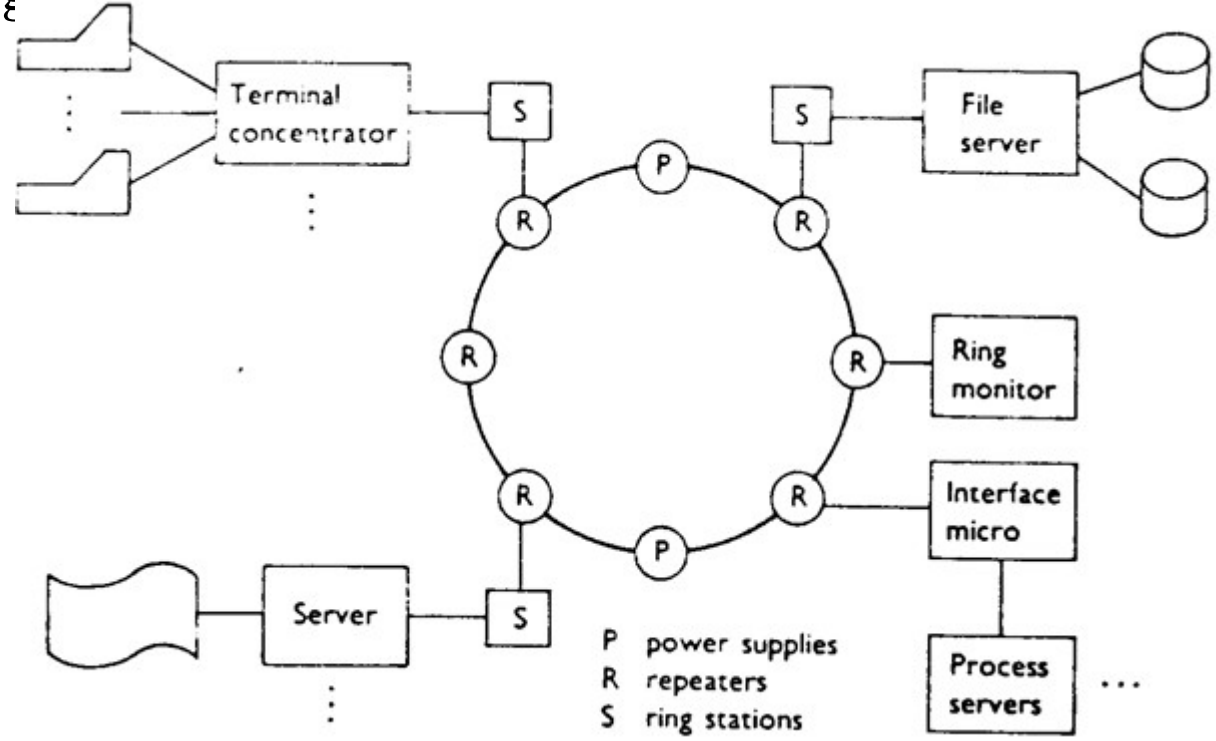
- Resource Sharing: number of resources, such as computers, peripherals, special-purpose processors, programs, databases etc. are interconnected by a communication system in order to allow the sharing of resources.
- These can be LANs (within an organization, research or industrial establishment) distributed and connected via WAN.
- Resource-sharing system may have overall common OS to be considered a homogeneous DCS or different OS for Heterogeneous DCS.

1. The Cambridge Distributed Computer System:

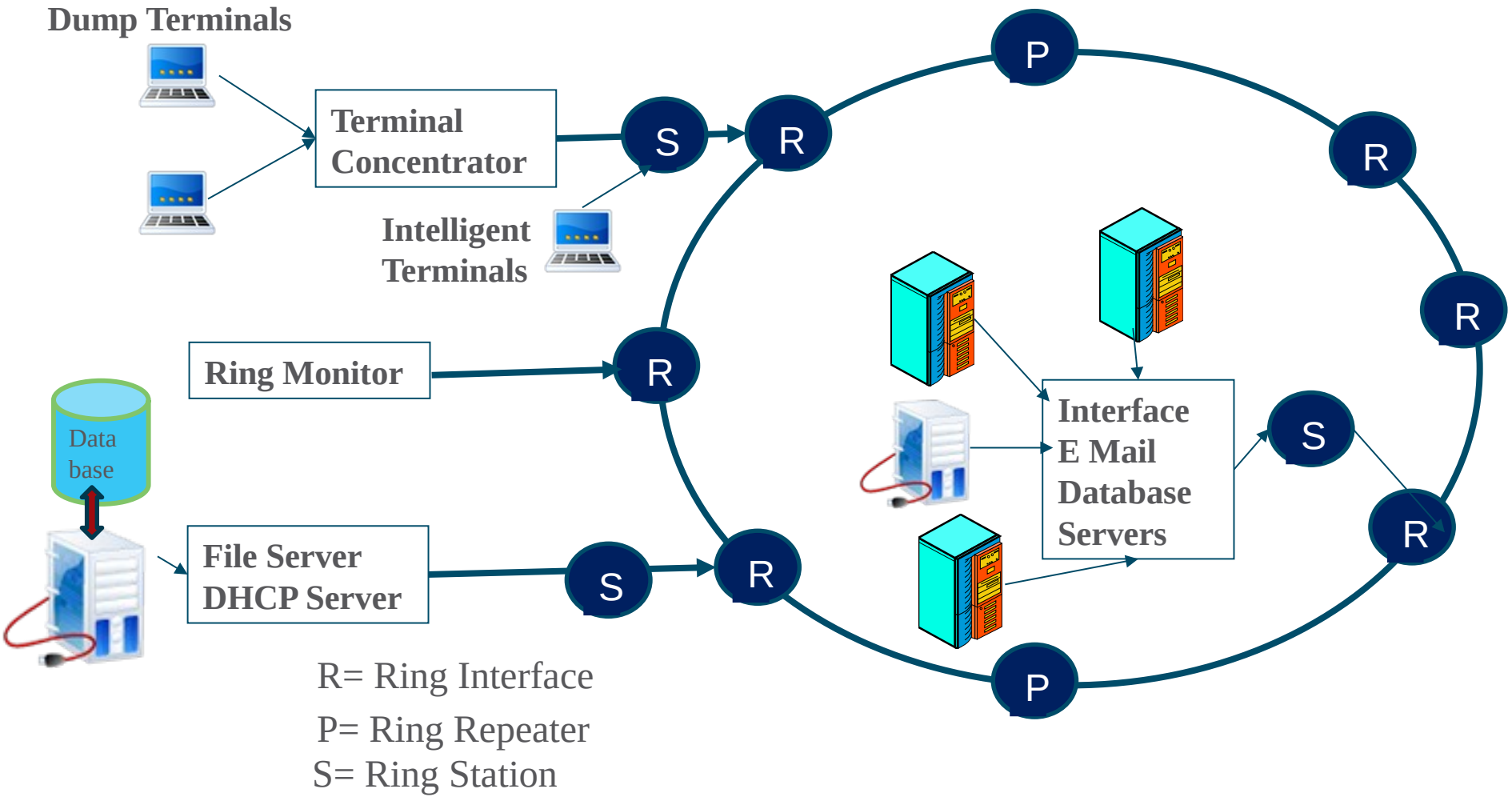
The system was designed to provide the services usually associated with a large conventional '**time-sharing**' system, except that it would employ small, self-contained PCs and terminals for user access, and mid size servers for those shared facilities (e.g. disks, printers, large memory and processing power).

The system components:

1. 50 heterogeneous machines.
2. Terminals connected via concentrators
3. 1 km Cambridge **fiber optics ring for communication**. This provides a train of fixed-size message slots endlessly circulating around the ring.



1. The Cambridge Distributed Computer System:



1. The Cambridge Distributed Computer System:

The Cambridge DCS resources sharing provides a good example of how DCSs manage key areas like security, email, and databases.

- **Security**: In a DCS system, data is spread across many servers, which makes security more challenging. The system needs to protect data and control access to it, using measures like encryption and secure login.
- **Email**: Managing email in a DCS involves handling how emails are sent, stored, and received across different servers. The system must ensure that emails are delivered reliably and that there are no errors in storage or routing.
- **Databases**: Distributed databases store data across multiple locations. The challenge is to keep the data consistent and available everywhere, even if some servers fail.
- The system needs to make sure data is synchronized and transactions are handled correctly.

Cambridge DCS had to solve these challenges to ensure smooth and secure operation in a distributed environment.

1. The Cambridge Distributed Computer System:

Cambridge DCSs Objectives: to provide a high-performance DCS system at low cost.

- The **loose coupling** between the system components would provides system **flexibility and resilience to failures**.
- The distributed nature of the system provides **transparent services.**
- The system consists of **heterogeneous components dedicated to perform particular function/service to the rest of the system.**
- DCS system provide **A4 (Authentication, Authorization, Accounting, and Auditing) services** to the distributed users.

1. The Cambridge Distributed Computer System:

- The software structure is based on the use of servers: the services of the system are provided by servers each performing a particular function.

Servers Used: Authentication

1)boot server: loads server code at initialization for diskless PC

- The Bootstrap Protocol (BOOTP) is a client/server protocol that configures a diskless computer or a computer that is booted for the first time. BOOTP provides:
  - IP address
  - net mask
  - the address of a default router
  - the address of a name server.



- BOOTP is static. When a client WS asks for the above information is retrieved from a fixed table. Every time the client asks for the info, it gets the same results.

1. The Cambridge Distributed Computer System:



- A **diskless node** (or **diskless workstation WS**) is a WS or PC without disk drives, which employs network booting by its NIC to load its OS from a BOOT Server using BOOTP. (A computer may said to act as a diskless node, if its disks are unused and network booting is used.)

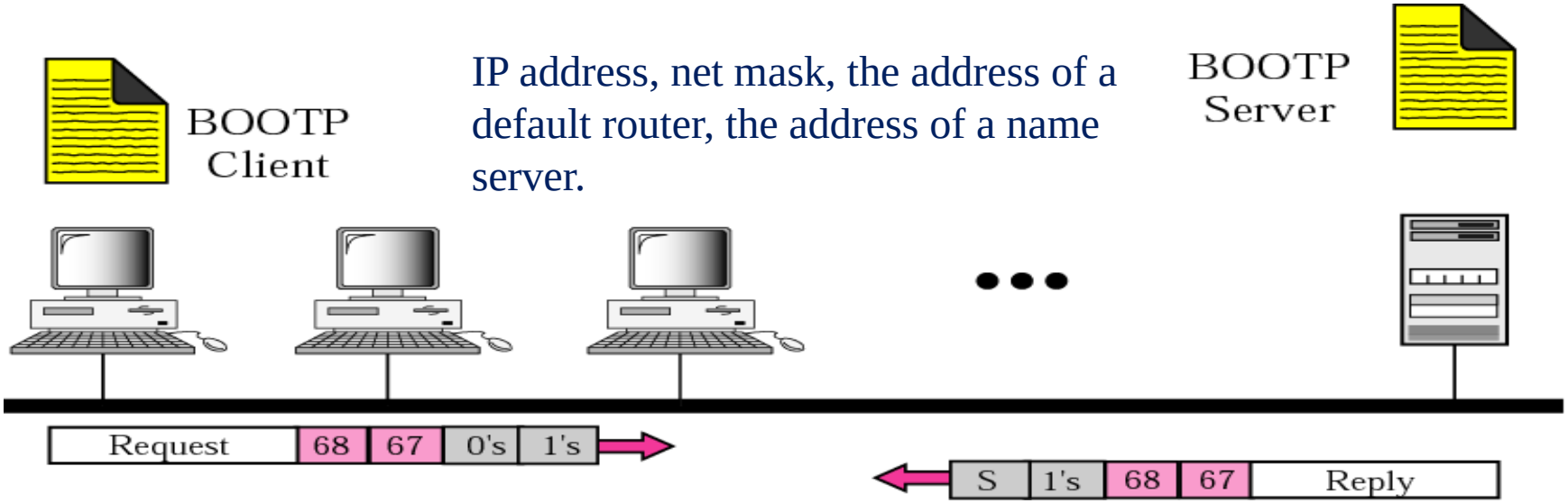
- Diskless nodes (or computers acting as such) are known as network computers or **hybrid clients**. Hybrid client may either just mean diskless node, or it may be used in a more particular sense to mean a diskless node which runs all applications remotely, as in thin client computing architecture.

- **Diskless PC:** Lacks local storage, relies on network for booting and operations.
- **Thin Client:** Low-end client designed to connect to a server or VDI for most operations, minimal local processing power.
- **Normal PC Workstation:** Standard personal computer with local resources, capable of independent operation.
- **Thick Client:** High-end client with substantial local resources and capability to operate independently of server interactions.

1. The Cambridge Distributed Computer System BOOT SERVER:

Case 1: Client and server on the same network

The BOOTP server can be on the same network as the BOOTP client or on different networks.



BOOTP places its packet inside a UDP packet (note that BOOTP is an application layer program).

1. The Cambridge Distributed Computer System BOOT SERVER:

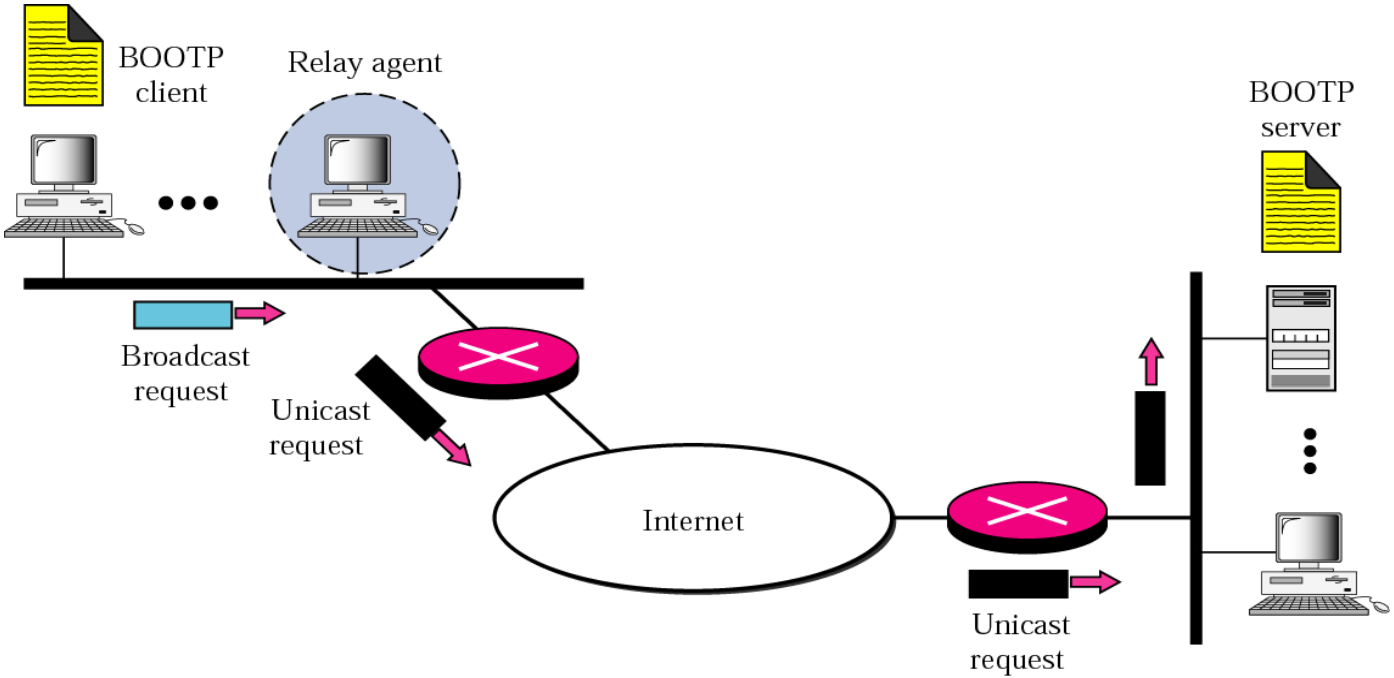
Case 1: Client and server on the same network

- The BOOTP server issues a passive open command on UDP port number 67 and waits for a client.
- A booted client issues an active open command on port number 68.
- The message is encapsulated in a UDP user datagram and then in an IP packet.
- In the IP packet the source address is all 0s and the destination address is all 1s.
- Server responds with a UDP datagram source port 67 and destination port 68.

1. The Cambridge Distributed Computer System BOOT SERVER:

Case 2: Client and server on two different networks

When client and server are on different networks, we need a relay agent, because client does not know IP address of server, and a limited broadcast address gets dumped by the local router. **Relay agent** knows the IP addr of the server.



1. The Cambridge Distributed Computer System BOOT SERVER:

Dynamic Host Configuration Protocol (DHCP)

- The Dynamic Host Configuration Protocol (DHCP) provides static and dynamic address allocation that can be manual or automatic.
- Bootp is static, but DHCP is dynamic (but it can also be static).
- DHCP has a pool of available addresses.
- When a request arrives, DHCP pulls out the next available address and assigns it to the client for a negotiable time period.
- When a request comes in from a client, the DHCP server first consults the static table.
- DHCP is great when devices and IP addresses change.

1. The Cambridge Distributed Computer System BOOT SERVER:

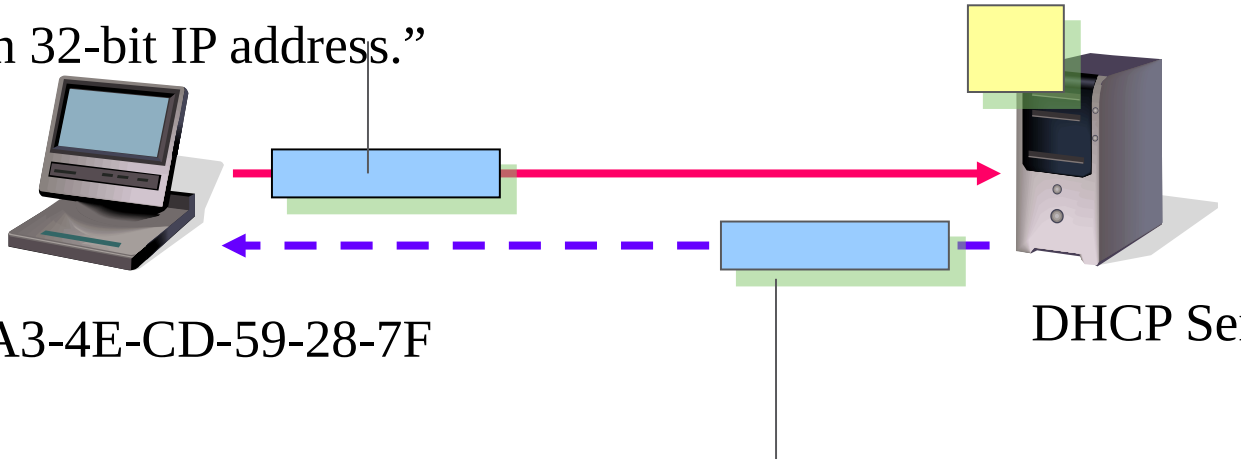
Dynamic Host Configuration Protocol (DHCP)

- DHCP used to provide IP configuration information to hosts on boot up.
- DHCP manages addressing by leasing the IP information to the hosts.
- The primary reason for using DHCP is to centralize IP management.
- When the DHCP service is used, DHCP scopes include pools of IP addresses that are assigned for automatic distribution to client computers on an as-needed basis, in the form of leases, which are periods of time for which the DHCP client may keep the configuration assignment.
- The address pools are centralized on the DHCP server, allowing all IP addresses on network to be administered from a single server.
- The next time the client machines are rebooted, they are assigned new addresses.

1. The Cambridge Distributed Computer System BOOT SERVER: Dynamic Host Configuration Protocol (DHCP)

1. DHCP Request Message:
"My 48-bit Ethernet address is A3-4E-CD-59-28-7F". Send me an 32-bit IP address."

2. Pool of IP Addresses



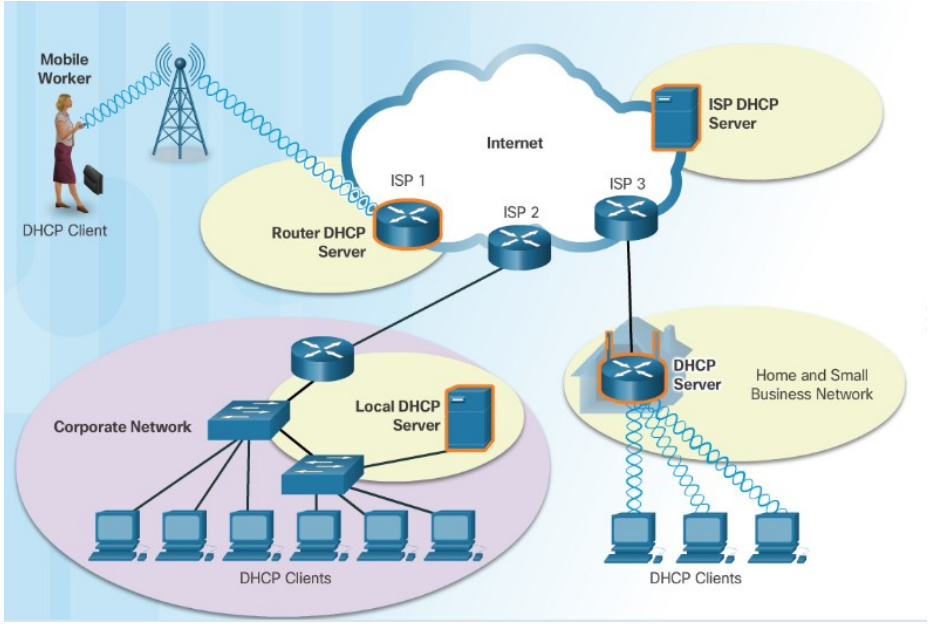
Client PC: A3-4E-CD-59-28-7F

DHCP Server

3. DHCP Response Message:
"Computer at A3-4E-CD-59-28-7F, your 32-bit IP address is 172.16.16.2
11010000101111101010101100000010".
(Usually other configuration parameters as well.)

1. The Cambridge Distributed Computer System BOOT SERVER: Dynamic Host Configuration Protocol (DHCP)

- DHCP automates the assignment of IPv4 addresses, subnet masks, gateways, and other parameters.
- DHCP-distributed addresses are leased for a set period of time, then returned to pool for reuse.
- DHCP is usually employed for end user devices.
- **Static addressing** is used for network devices, such as gateways, switches, servers, and printers.



DHCP Information can include:
•IP address, Subnet mask, Default gateway, Domain name, DNS Server.

1. The Cambridge Distributed Computer System:

Servers Used:

2) User authentication server (provides security **tokens** or **Smart Cards** to prevent resource usage by an unauthorized user):

- User authentication and protection was essential to prevent unauthorized access.

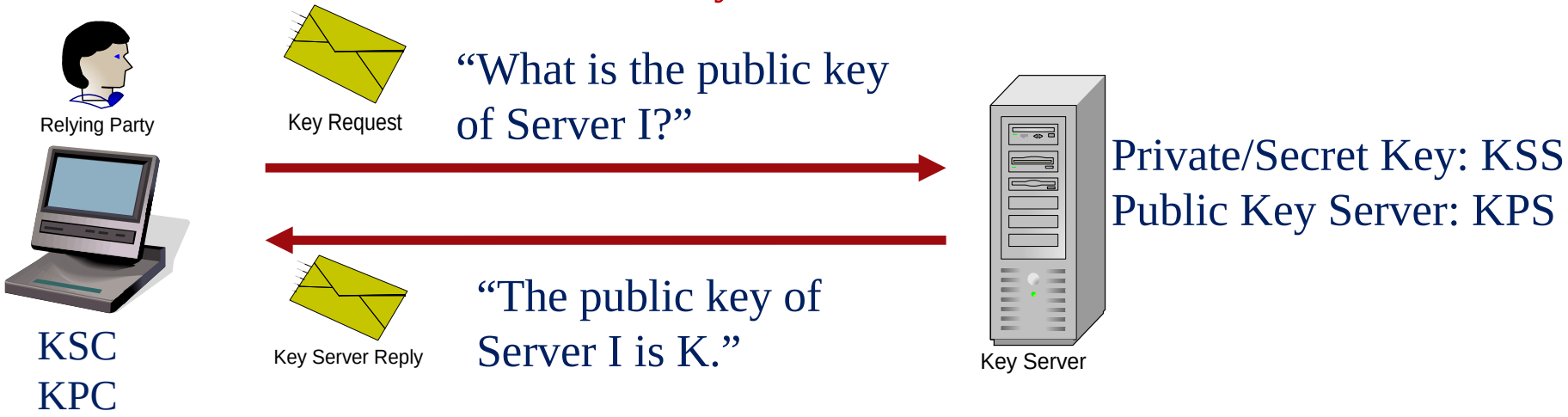






1. The Cambridge Distributed Computer System:

Authentication server: KDC Key Distribution Server



- The key server stores [identity, public key] pairs of all servers (application, dbase,…)
- The key request can be in plaintext
- The key server reply is encrypted using the private key KSS of the key server and the public key KPC of the client C
- The public key KPS of key server is known to the relying party (Client C)
- The Client decrypts the reply using Server Public key KPS and Client Secret key KSC

1. The Cambridge Distributed Computer System:

3) Processing servers (processor bank):

- For work which is too demanding for a user's local machine, requests can be made to the processing servers.
- According to the requirements, allocation will be made to one of the heterogeneous collection of minicomputers which support different OSs.
- It is possible to run concurrent programs by the allocation of multiple machines.

1. The Cambridge Distributed Computer System:

4) Resource manager:

- This server keeps account of what has been allocated, to whom, and for how long. It is responsible for allocating and reclaiming resources.

5) File server:

- This provides a uniform interface accessible from all the different machine subsystems.
- All the required functions are provided, such as disk management and integrity, atomic updates and interlocks for shared access.
- It has a basic naming structure using unique identifiers which prevents unauthorized access and allows superimpose different data structures.

1. The Cambridge Distributed Computer System:

6) Time stamp server (time of day):

provides real time stamp to messages handled by the system users.

7) Name server (DNS): gives the location of a named service or machine.

8) Message server:

exchange of users mails and also used for technical support to provide adequate user response and system performance.

1. The Cambridge Distributed Computer System:

9) Network Management

- Networks and distributed processing systems are of critical and growing importance in business, government, and other organizations.
- Within a given organization, the trend is toward larger, more complex networks and distributed processing systems supporting more applications and more users.
- A large, reliable network cannot be put together and managed by human effort alone.
- The complexity of such a system dictates the use of **automated network management tools**.
- The urgency of the need for such tools is increased, and the difficulty of supplying such tools is also increased, if the network includes equipment from multiple vendors.
- In response, standards that deal with network management have been developed, covering services, protocols, and management information base.

1. The Cambridge Distributed Computer System:

9) Simple Network Management Protocol (SNMP):

- SNMP allows network administrators to collect information about the network.
- It is a communications protocol for collecting information about devices on the network, including hubs, routers, and bridges.
- Each piece of information to be collected about a device is defined in a Management Information Base (MIB).
- SNMP uses UDP to send and receive messages on the network.

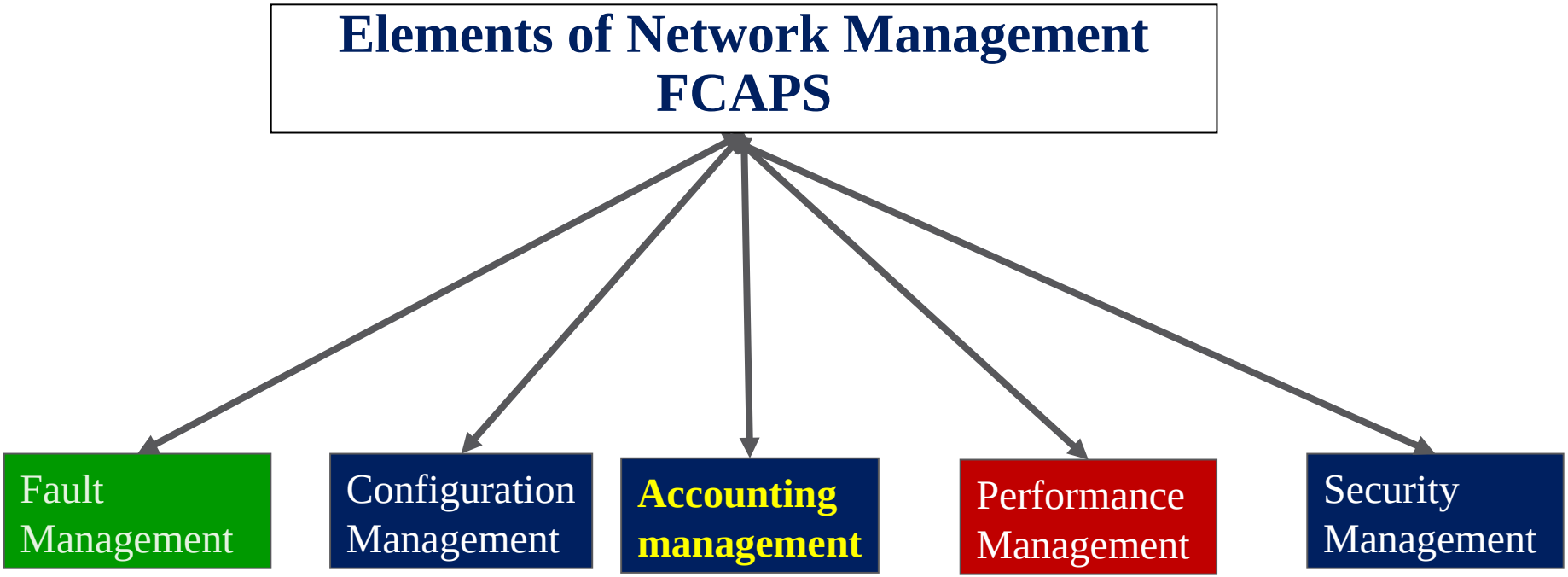
1. The Cambridge Distributed Computer System:

9) Network Management System NMS

NMS is a collection of tools for network monitoring and control that is integrated in the following senses.

- It has a single operator interface with a powerful but user-friendly set of commands for performing most or all network management tasks.
- It needs a minimal amount of additional equipment.
- Most of the hardware and software required for network management is incorporated into the existing user equipment.
- A network management system consists of incremental hardware and software additions implemented among existing network components.
- The software used in accomplishing the network management tasks resides in the host computers and communications processors (e.g., switches, routers).

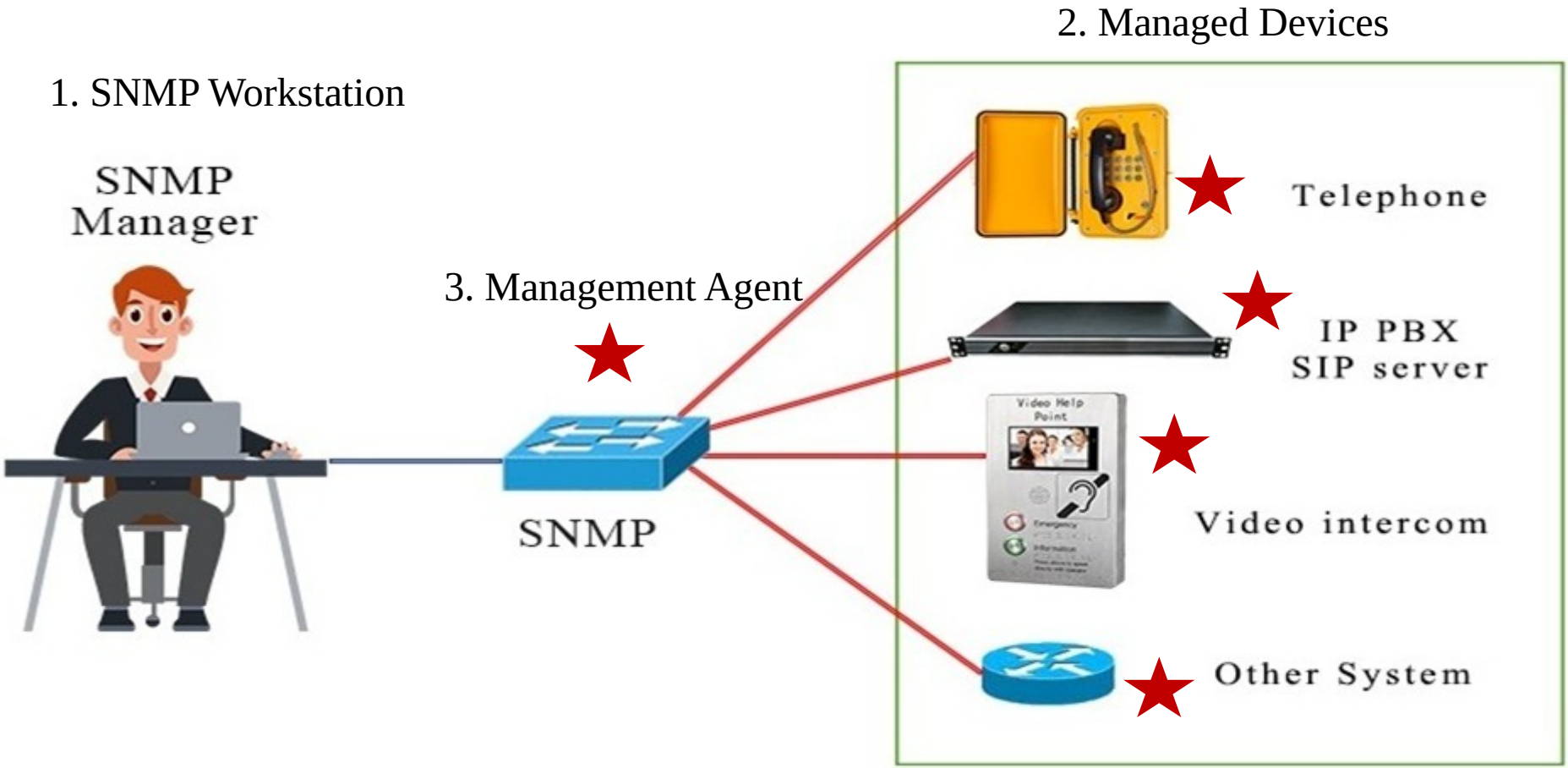
Elements of Network Management Systems NMS



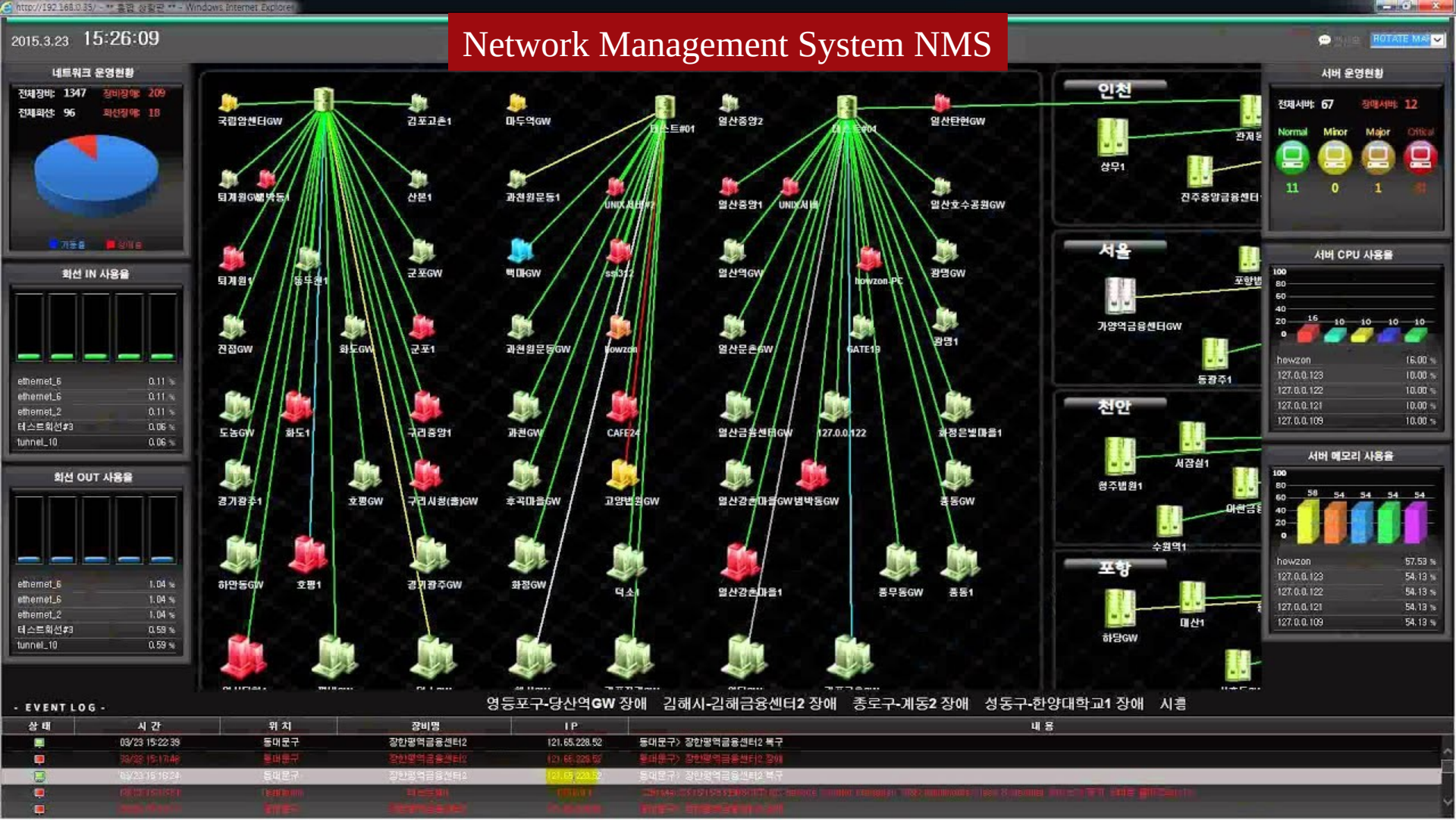
Elements of Network Management FCAPS

1. **Fault Management FM:** set of functions that detect, isolate, and correct malfunctions in a network, compensate for environmental changes, and include maintaining and examining error logs, accepting and acting on error detection notifications, tracing and identifying faults, carrying out sequences of diagnostics tests, correcting faults, reporting error conditions, and localizing and tracing faults by examining and manipulating database information.
2. **Configuration Management CM:** focuses on establishing and maintaining consistency of a system or product's performance and its functional and physical attributes with its requirements, design, and operational information throughout its life.
3. **Accounting Management AM:** techniques to control and report on the financial health of the organization. This involves the analysis, planning, implementation, and control of programs designed to provide financial data reporting for managerial decision making and performance analysis.
4. **Performance Management PM:** activities that ensure that goals are consistently being met in an effective and efficient manner. PM focus on the performance of an organization, a department, employee, or even the processes to build a product or service, as well as many other areas.
5. **Security Management SM:** related to asset management, physical security and human resource safety functions. It entails the identification of an organization's information assets and the development, documentation and implementation of policies, standards, procedures and guidelines.

Network Management System NMS



Network Management System NMS



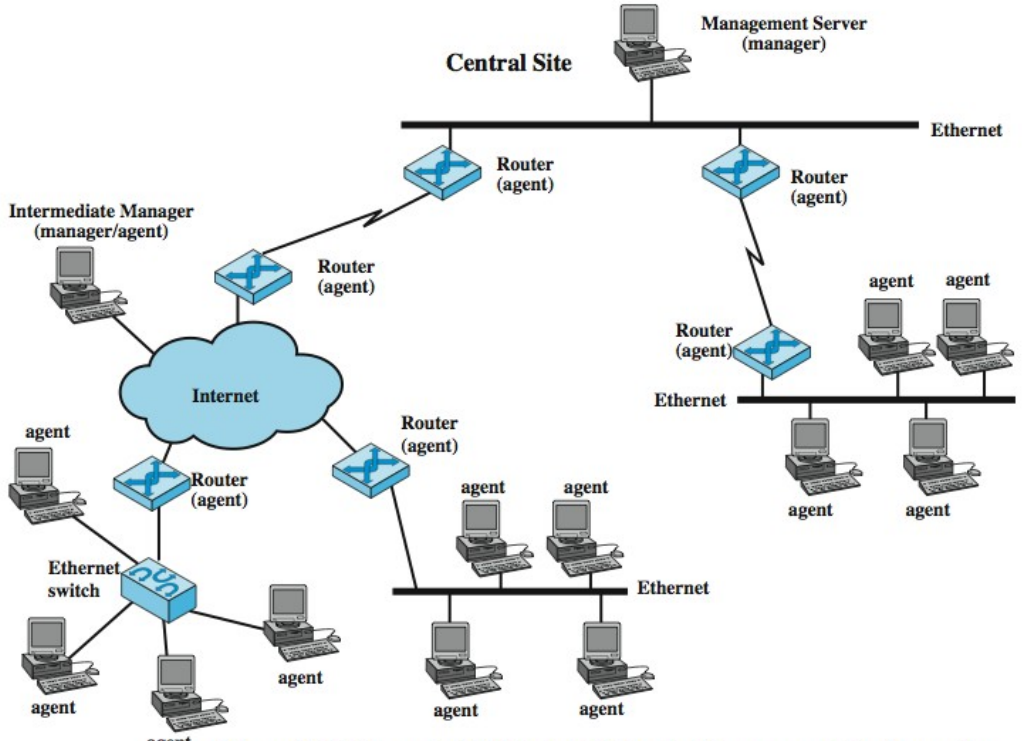
1. The Cambridge Distributed Computer System:

9) Network Management

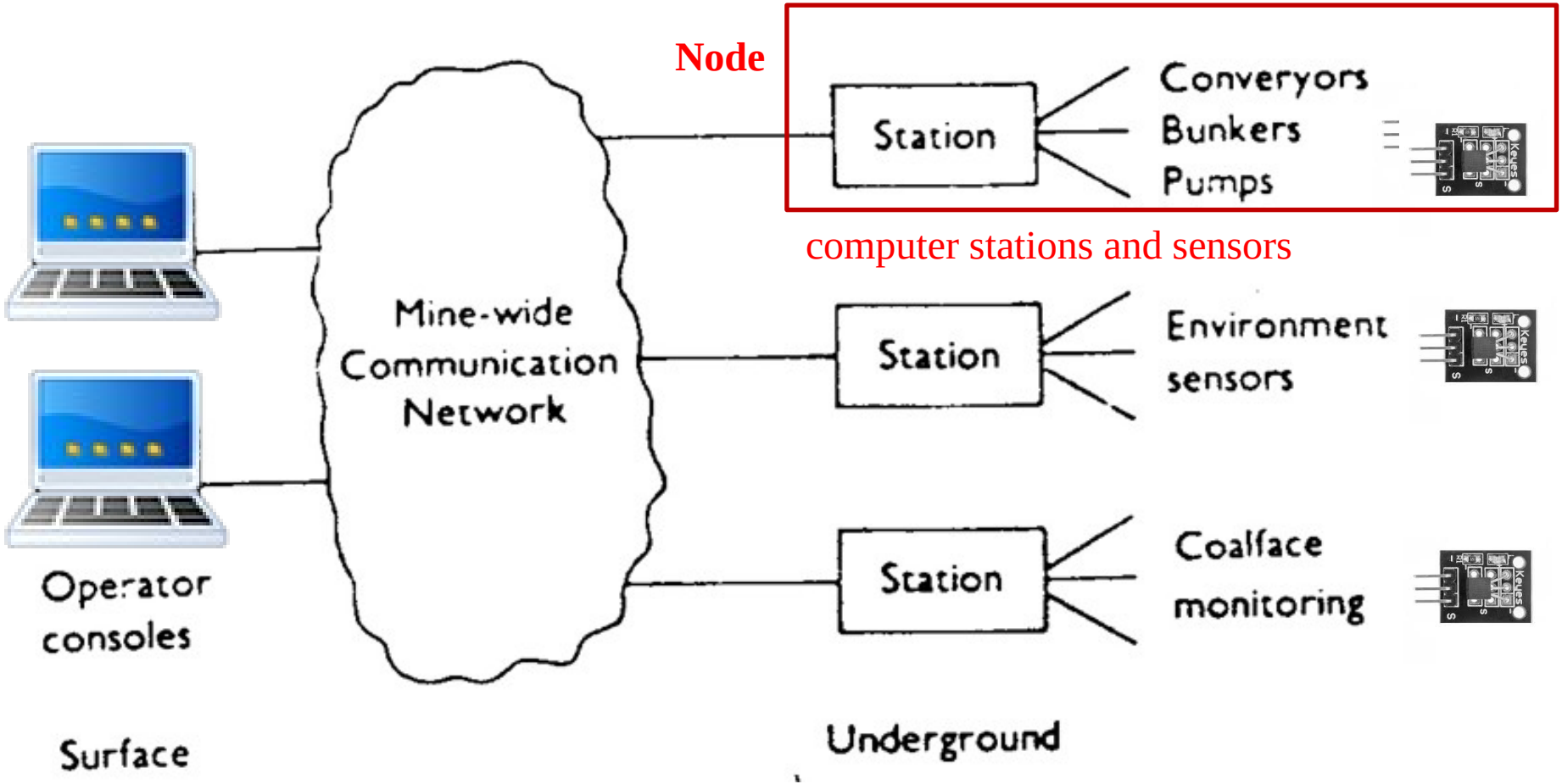
The SNMP includes the following key elements:

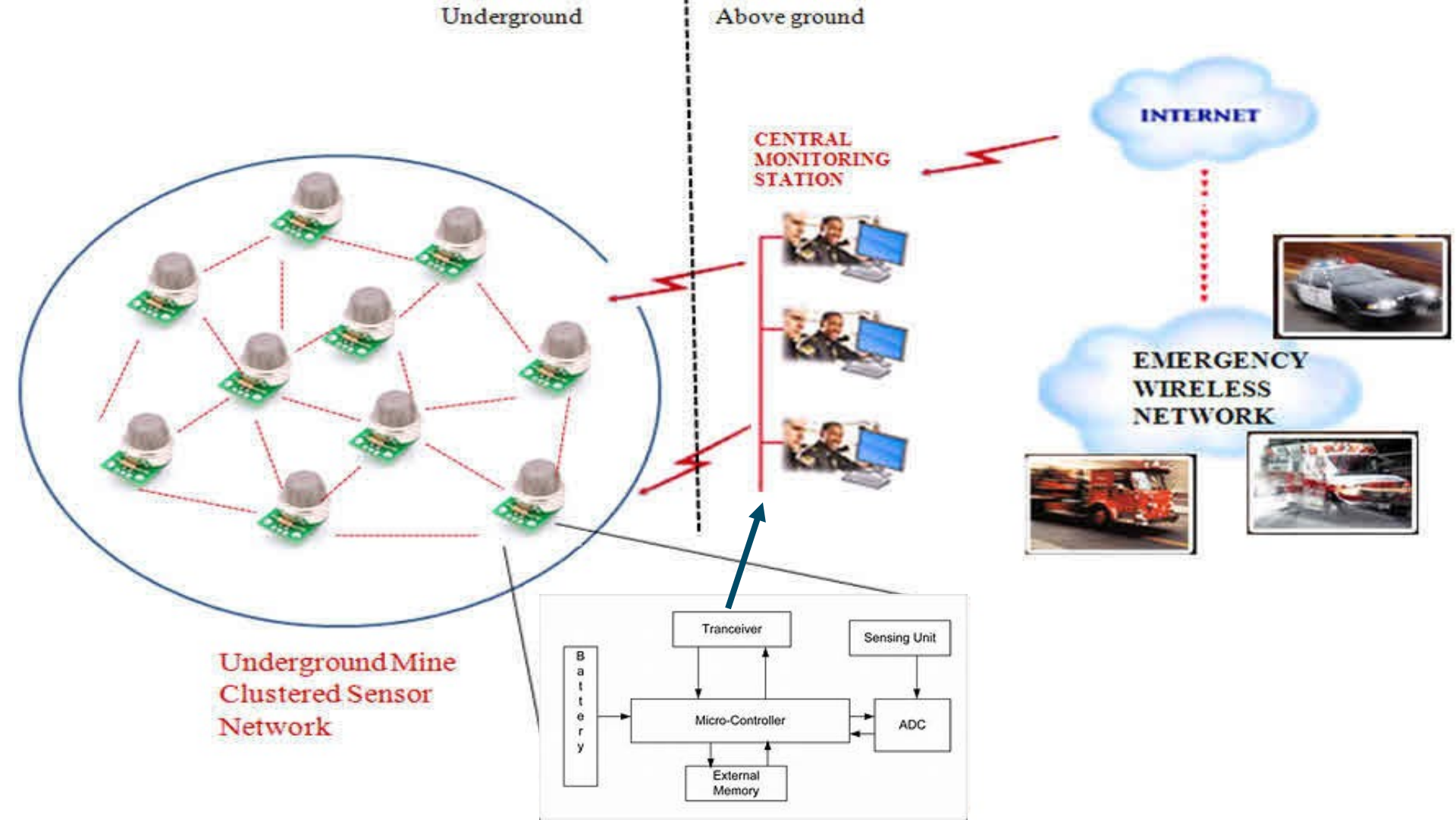
- Management station, or manager
- Agent
- Management information base
- Network management protocol

The SNMP includes the following functions: Alarm Monitoring, Configuration Management, Fault management, and Security Management



2. DCS for underground monitoring and control in coal mines:



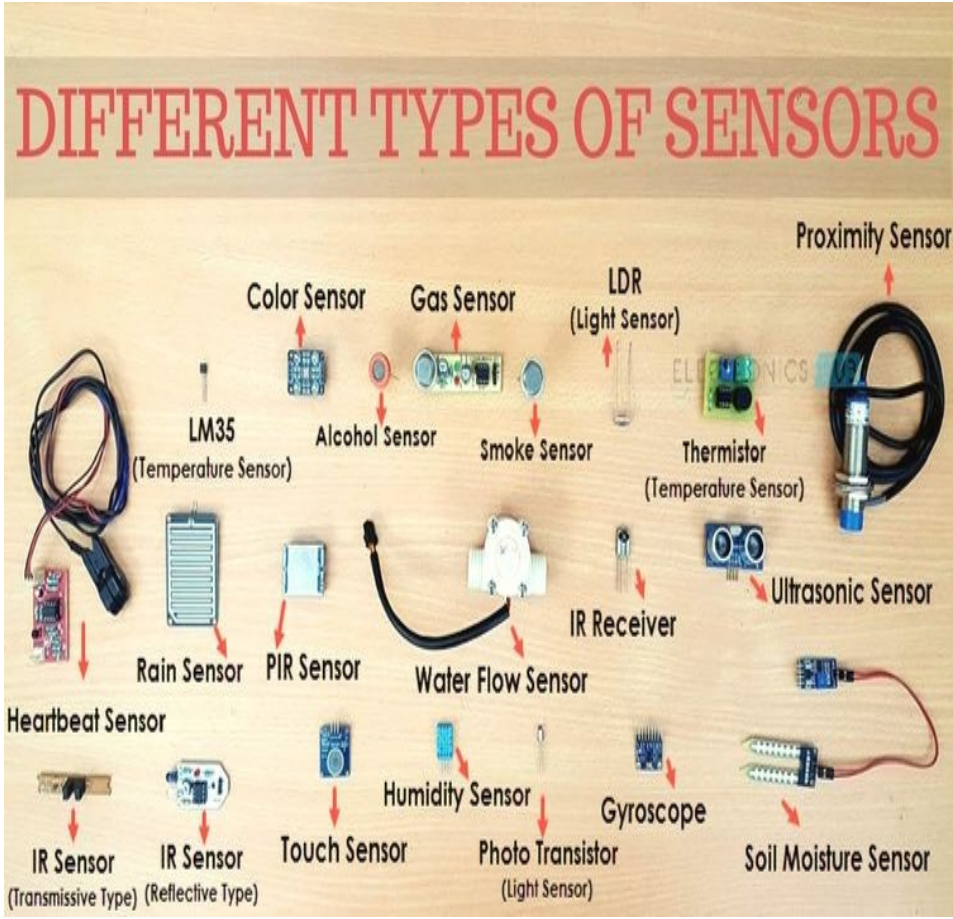

Underground
Above ground
INTERNET
CENTRAL MONITORING STATION
EMERGENCY WIRELESS NETWORK
Underground Mine Clustered Sensor Network
Tranceiver
Sensing Unit
Battery
Micro-Controller
ADC
External Memory

2. Process Control and Laboratory Automation in coal mines :

- There are a large number of diverse activities in a coal mine which would greatly benefit from computer monitoring and control: coal cutting and extraction, coal transport, power consumption, monitoring of the environment and control of fans and water drainage.
- The functions required vary from simple remote monitoring to full automatic control.
- **Automation in coal mines** provide a **safer environment** by more comprehensive monitoring and enabling personnel to be removed or remote from some of **the dangerous and unhealthy activities.**
- The main characteristics of such a system are that it would be very large (hundreds of computer stations and sensors), physically distributed (>=1 km), and real-time.

2. Process Control and Laboratory Automation in coal mines :

- multiple microcomputers are used as sensors performing the local monitoring and control.
- These are interconnected (via a master computer) to coordinate their actions.
- A device station would thus include the device **(a sensor, actuator or controller)** directly interfaced to a processor and provided with a communications interface for direct connection onto a network.
- Each device station communicates with other stations, and possibly a central computer, to cooperate in the overall control of the plant.



Different Types of Sensors



Thermistor
(Temperature Sensor)



IR Sensor
(Transmissive Type)



IR Sensor
(Reflective Type)



Ultrasonic Sensor



Gyroscope
Sensor



Accelerometer
Sensor



Rain Sensor



Soil Moisture Sensor



Phototransistor
(Light Sensor)



Water Flow Sensor



Heartbeat Sensor



Alcohol Sensor



Color Sensor



PIR Sensor



Gas Sensor



Smoke Sensor



LM35 (Temperature
Sensor)



IR
Receiver

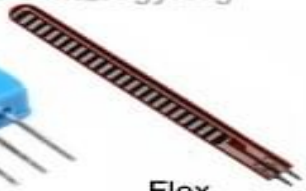


LDR (Light
Sensor)

www.electricaltechnology.org



Humidity Sensor



Flex
Sensor



Touch Sensor



Solar Cell
Light Sensor



Metal
Dedector



Real Time
Clock Sensor



Vibration
Sensor

2. Process Control and Laboratory Automation in coal mines :

Advantages of the use of local processors/sensors are:

1) Reduction in wiring costs:

- Industrial sites usually extend over distances of up to 1km.
- Wiring can often account for a substantial proportion of the cost of a control system.
- **Wireless or Serial transmission lines** can be used rather than the traditional use of large numbers of parallel lines linking instruments with a central computer.

2) Faster response

- Response time is one of the most important factors in the design of the control system.
- A local processor can be dedicated to a device and so respond very quickly.
- communication system for process control application differs from that for other applications in that **low delays and response times** are more important than **high throughput**.

2. Process Control and Laboratory Automation in coal mines :

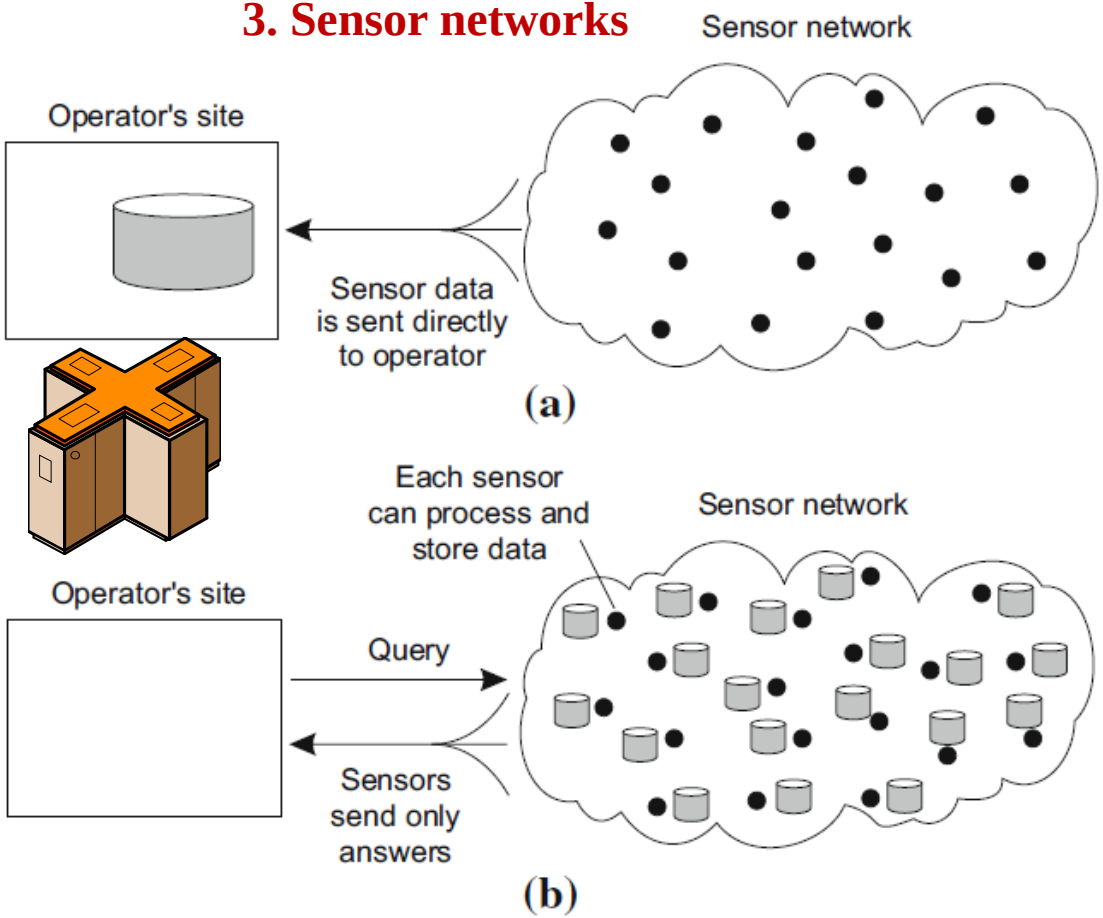
Advantages of the use of local processors are:

3) Easier development and maintenance:

- This is aided by the modularity of the software.
- the local intelligence in the device means it can be isolated from the rest of the plant for development or maintenance purposes.

4) Degraded operation: local processing can continue to provide some level of control even when the central control station has failed.

3. Sensor networks



Sensing:

- Temperature, humidity, (vehicular) movement, lightning condition, pressure, soil make-up, noise Levels
- Presence or absence of certain kinds of objects
- Mechanical stress levels on attached objects
- Current characteristics such as **speed, direction, and size of an object**

Fig. Organizing a sensor network database, while storing and processing data: a) only at the operator's site or b) only at the sensors

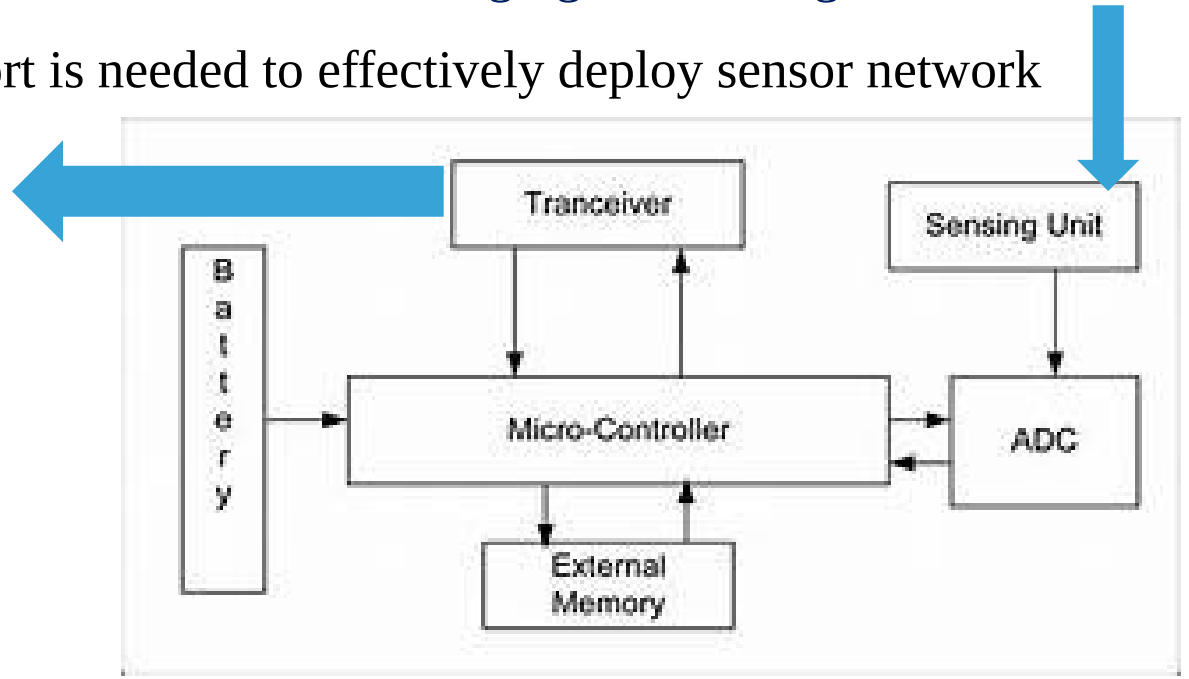
3. Sensor Networks SN, WSN

- An **actuator** is a mover component of a machine responsible for moving and controlling a mechanism or system, for example by opening a valve.
- An actuator requires a control signal and a source of energy.
- The control signal is relatively low energy and may be electric voltage or current, pressure, or even human power.
- Its main energy source may be an electric current, or pressure.
- When it receives a control signal, an actuator responds by converting the source's energy into mechanical motion.
- An **actuator is a mechanism** by which a control system acts upon to perform an operation or task.
- The control system can be simple (a fixed mechanical or electronic system), software -based (e.g. a printer driver, robot control system, a human, or any other input).

3. Sensor Networks SN, WSN

- **Sensor Node architecture:** above the hardware there is a software layer, low-level network access, access to sensors and actuators, memory management, and so on.
- support for specific services is included, such as localization, local storage/flash devices, and communication facilities such as messaging and routing.
- Additional middleware support is needed to effectively deploy sensor network applications.

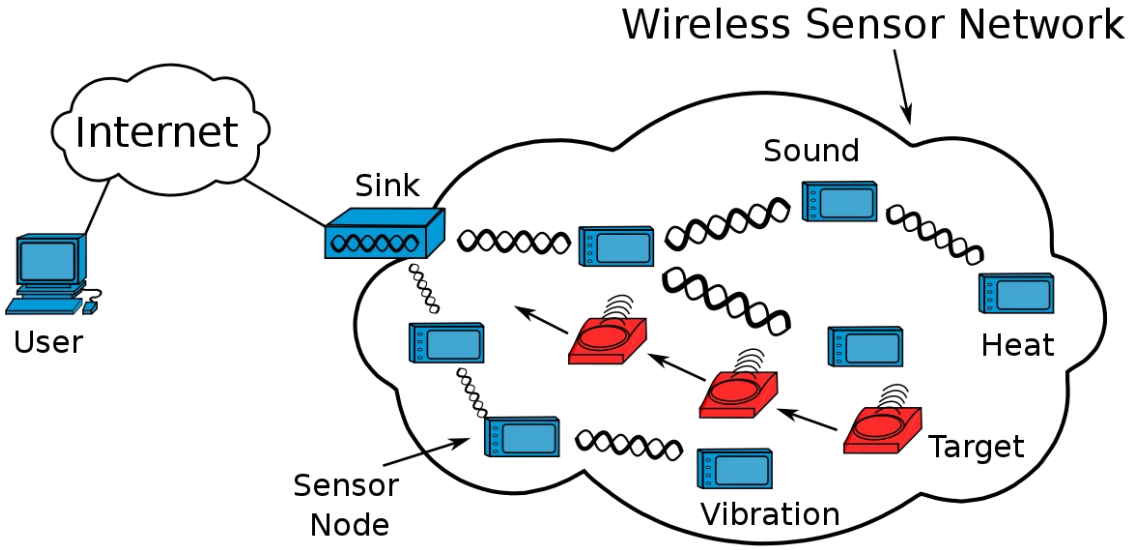




3. Sensor Networks SN, WSN

Sensors Types:

1. Chemical sensor
2. Biosensor (biomedicine and biotechnology)
3. MOS Metal Oxide Semiconductor sensors
4. Biochemical sensors
5. Image sensors





Ultrasonic sensor

3. Sensor Networks SN, WSN applications

- Industrial control and automation
- Security and public safety
- Agriculture monitoring
- Environmental monitoring
- Health applications
- Smart space/home applications
- Traffic Control, vehicle tracking and detection
- Interactive museums
- Infrastructure security, military applications



3. Sensor Networks and IOT Applications

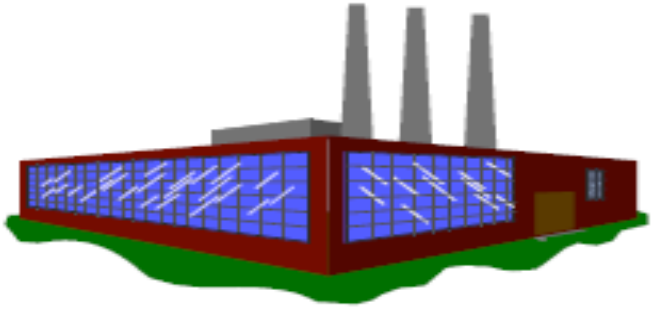
Smart Spaces



Smart School



Smart City



Smart Factory

Other Applications

- Battlefields/Surveillance
- Earthquake areas
- Environmental Monitoring
- Airport security
- Emergency Response
- Location Services

4. The Bank of America distributed system:

- Bank of America in California operated a very large database consisting of information on their customer's accounts - 10 million of them!
- The bank had over 1000 community offices and two main centers, one in Los Angeles and one in San Francisco.
- During office hours an estimated peak of 50 account enquiries/second per center were made.
- The overnight hours are used to perform the total of approximately 4 million transactions, transits and clearing items to other banks.
- The DCS consists of two supercomputers and databases at the centers and have multi-drop communication lines to the community offices and a communication line between the centers.

4. Bank of America Mainframe Architecture

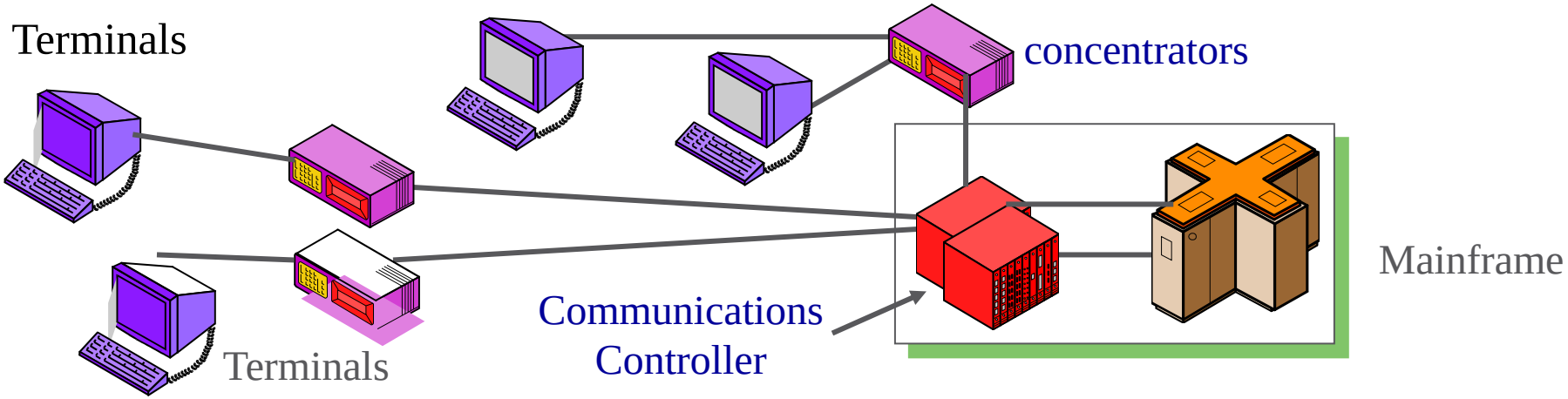
1. Central Mainframe

2. Communications Controllers: Front End Processors

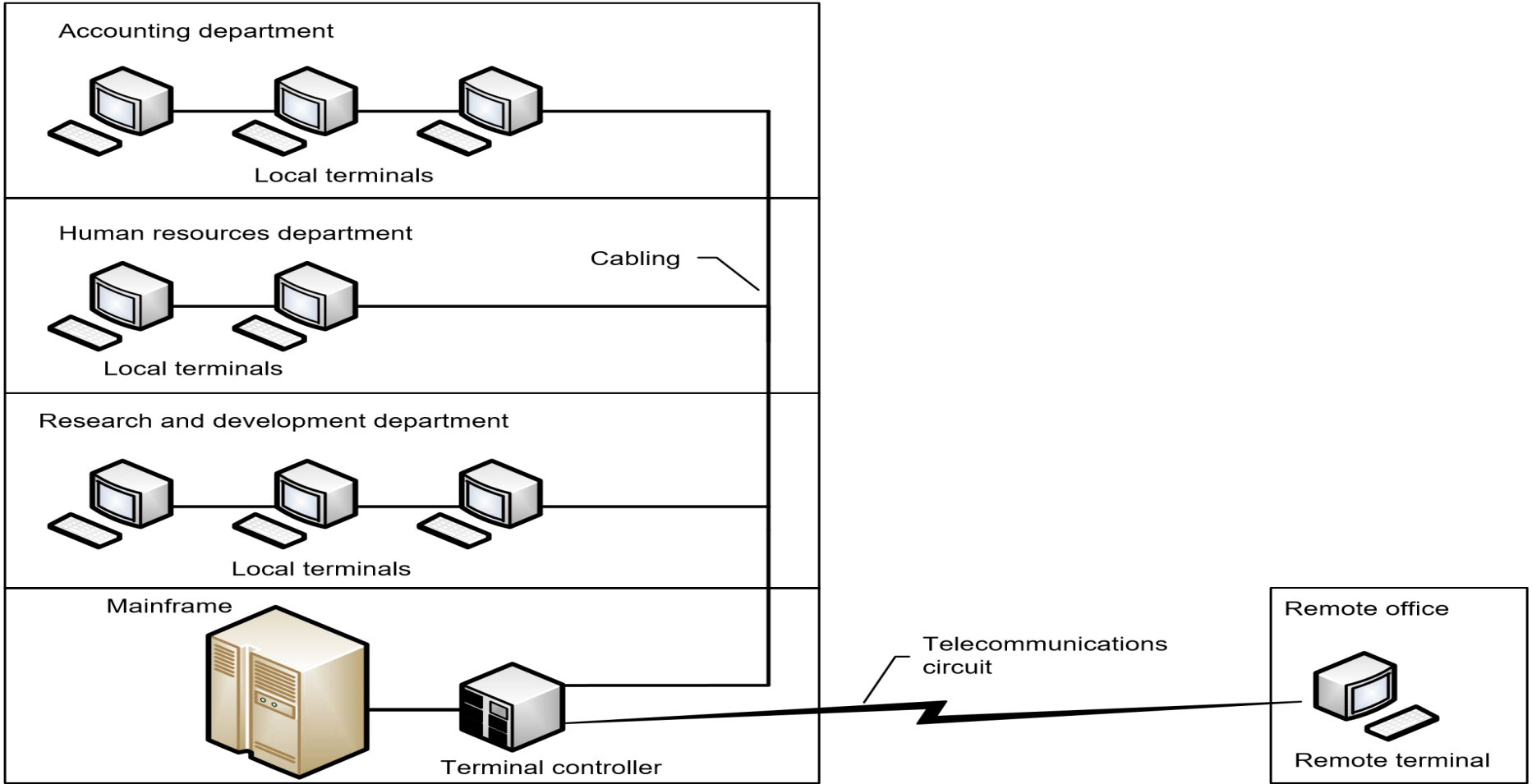
- Handles multiplexing to reduce transmission cost
- Handles detailed interactions with cluster controllers, freeing mainframe to deal with database processing

3- Terminal concentrators/multiplexors/cluster controller

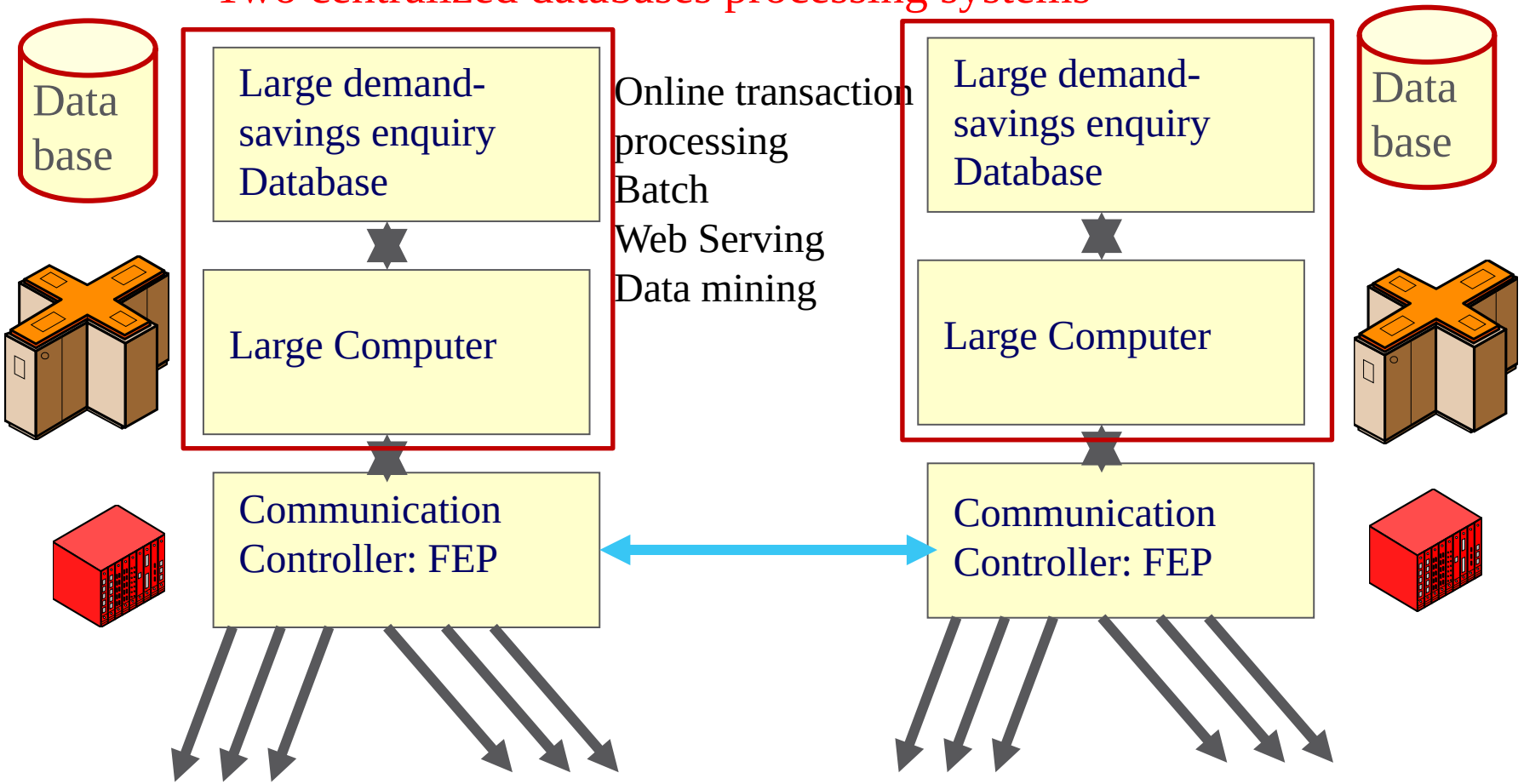
4- Terminals



4. Bank of America Mainframe Architecture



4. The Bank of America distributed system
Two centralized databases processing systems



4. The Bank of America distributed system:

Architecture features:

- Reliability: Use expensive Standby computers
- Performance: Response for queries and throughput for batch and on line processing are sufficient (low response time and high throughput)
- Extension and upgrading: New larger central computer or upgrade existing one
- Maintenance: Of both the hardware and software using redundancy in the system.

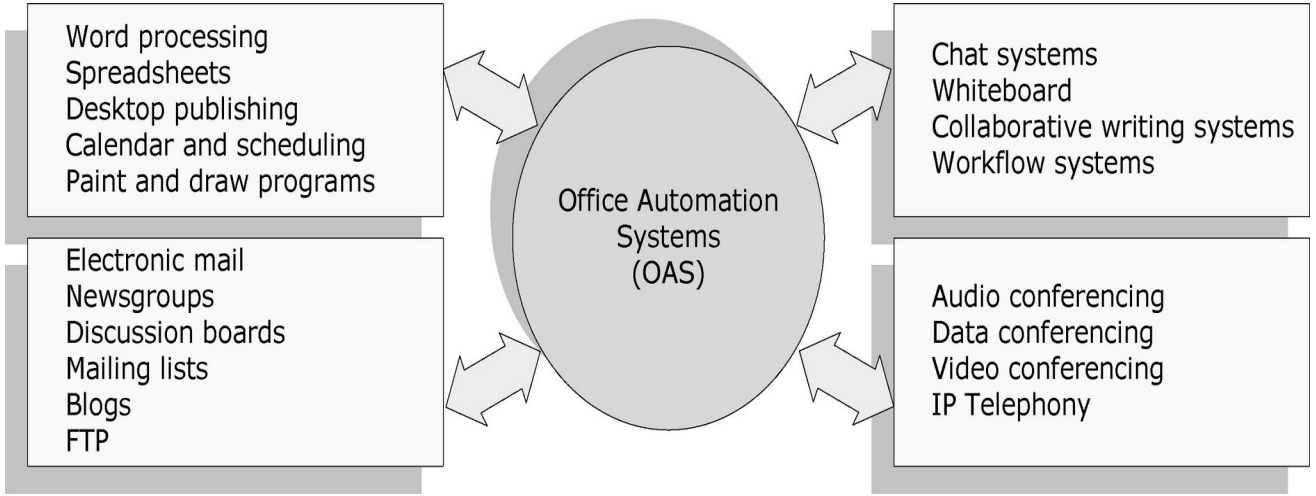
4. The Bank of America DS

- All transactions were logged for overnight processing and data base update.
- They avoided many of the difficult problems of consistency in replicated data bases in an environment of concurrent, real-time access and update.
- The Bank of America system is implemented using IBM general purpose systems to support transaction processing between autonomous computer systems.
- DBMS is IMS (Information Management System) which coordinates transactions dealing with data base enquiries and updates, and integrates with a system for inter-program communication SNA, Systems Network Architecture products of IBM.

5- Office Automation System AOS

Office automation is one of the fastest-growing applications of resource sharing including:

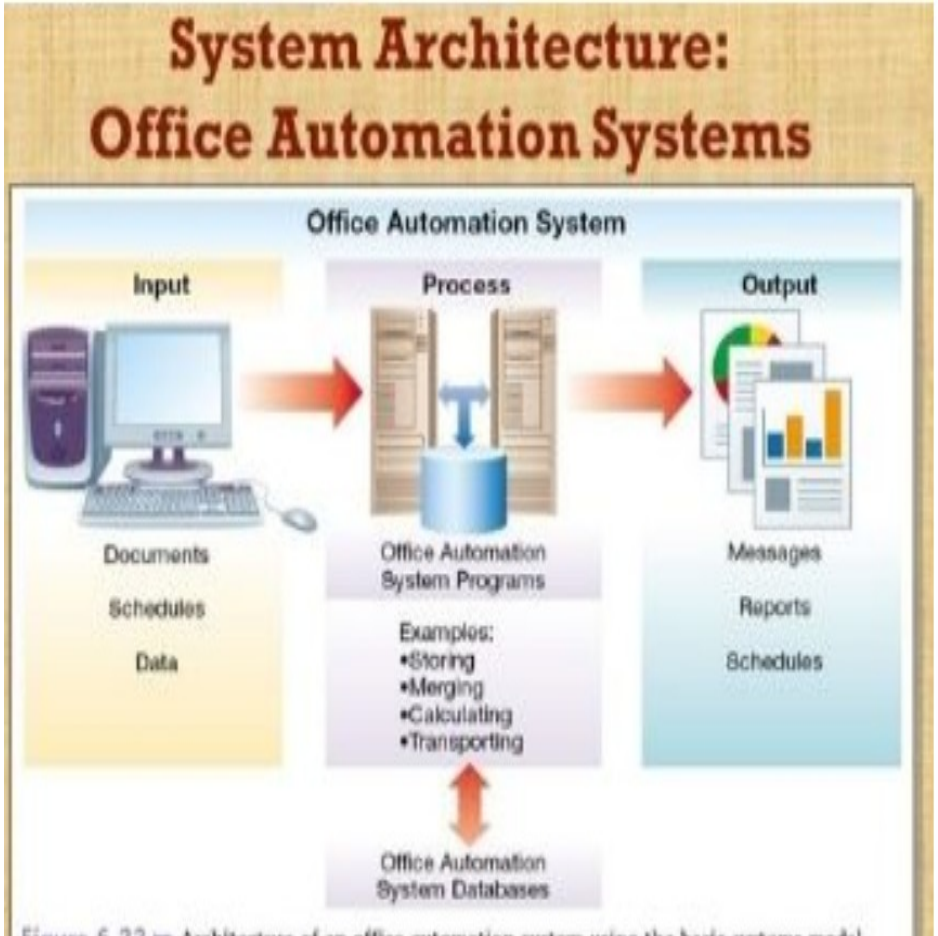
1. E-mail, document filing, conferencing, and resource sharing, such as laser printers and databases.
2. Intelligent terminals for word processing, accounting, some graphics etc., office automation systems provide other services.



Office Automation System AOS

Office Automation Systems (OAS) are distributed CBIS that are used to execute a variety of office operations, such as word processing, electronic spreadsheet, e-mail, and video conferencing.

These different OAS allow the automation of much of the administrative work in the office and typically focuses on the more repeatable and predictable aspects of individual and group work.



Office Automation System AOS

AOS are more and more frequently used by managers, engineers, and clerical employees to increase efficiency and productivity.

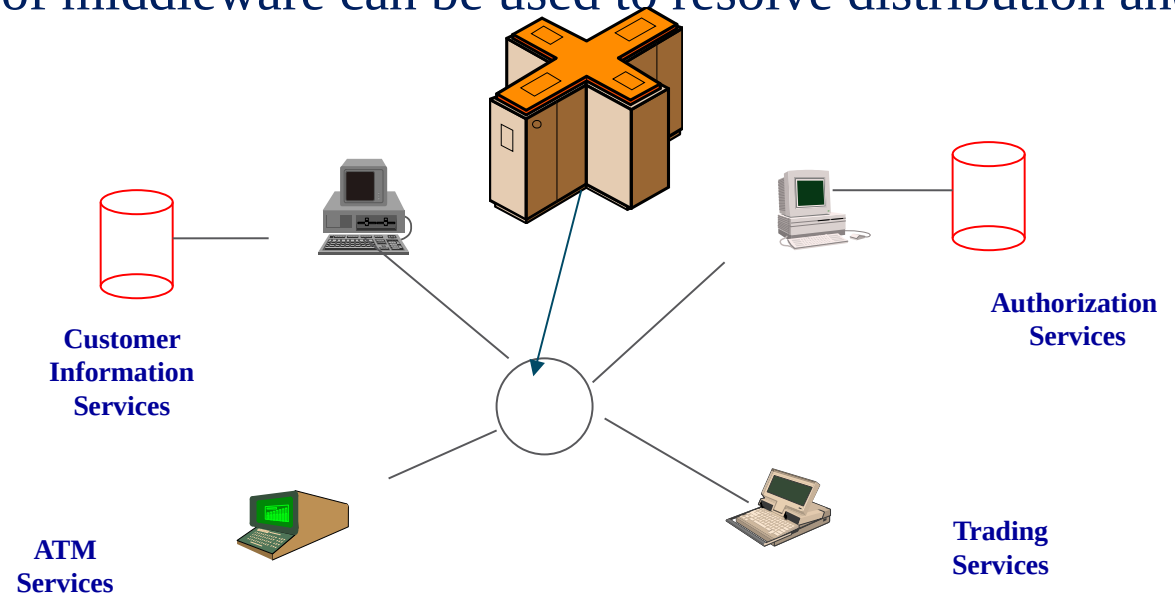
They support the general activities of workers and underlie the automation of document-centric tasks performed by production office workers.

The OAS encompass a broad set of capabilities, and provide much of the technological basis for the electronic workplace.

The focus of OAS have typically been used in supporting the information and communication needs of office workers, and its use by organizations supporting the white-collar work force has revealed itself crucial..

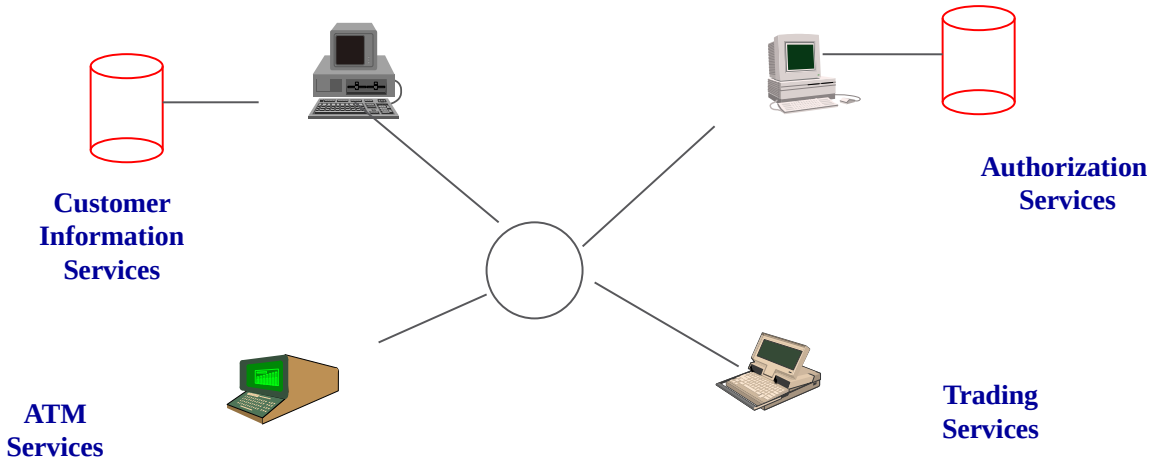
6- IT Service Architecture of a Bank

- Service architecture consists of heterogeneous new and legacy components
- Hardware platforms range from mainframes to WINTEL
- Programming languages including Assembler, C++, Java, …
- Different types of middleware can be used to resolve distribution and heterogeneity

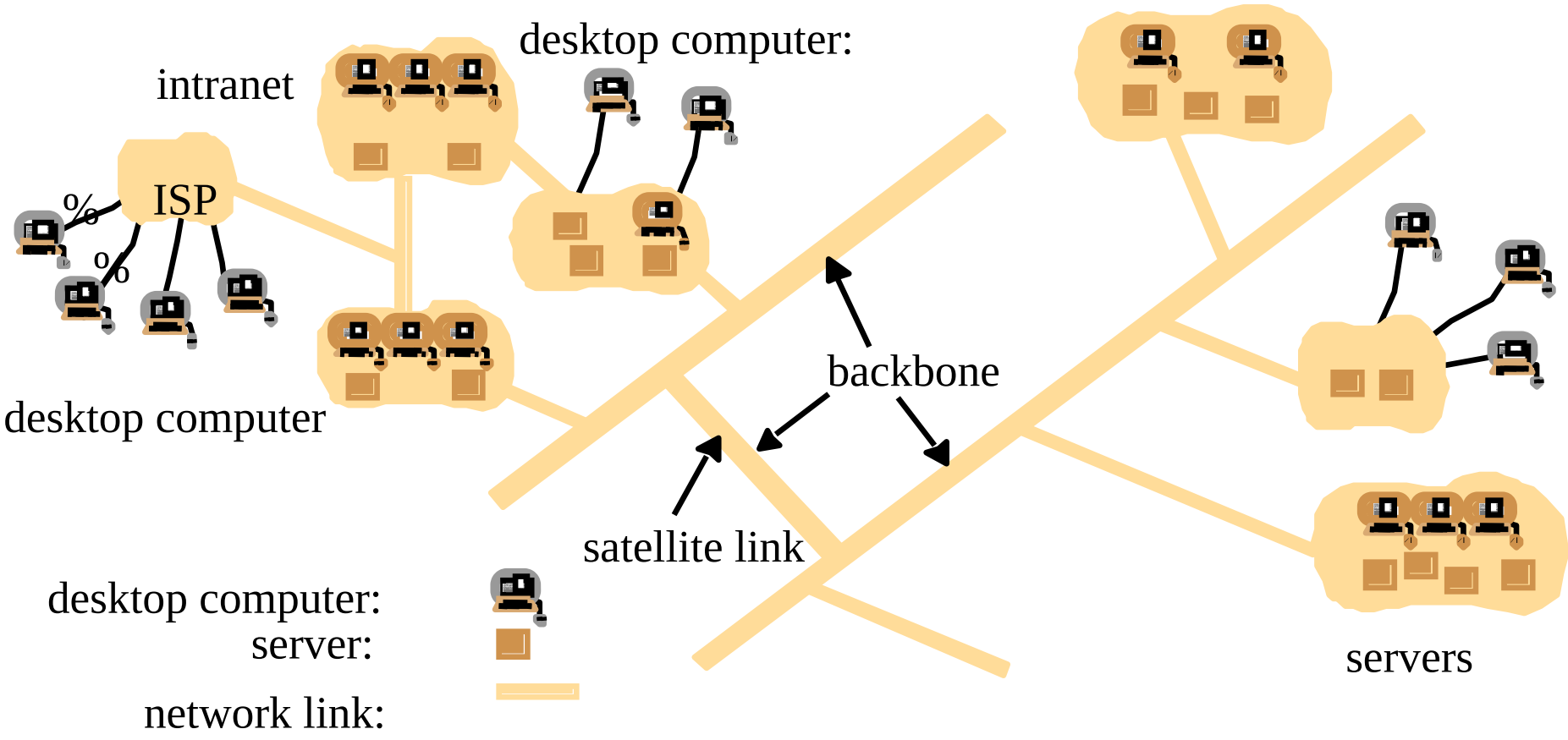


IT Service Architecture of a Bank

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7. The Internet and WWW



7. The Internet and WWW

- The Internet is a vast collection of computer networks of many different types and hosts with various types of services.
- Programs running on the computers connected to it interact by passing messages, employing TCP/IP Protocols.
- It enables users, wherever they are, to make use of open-ended services (WWW, email, file transfer, multimedia services, ...)
- World Wide Web is the largest software application running on Internet. It becomes the most popular distributed software application ever created.



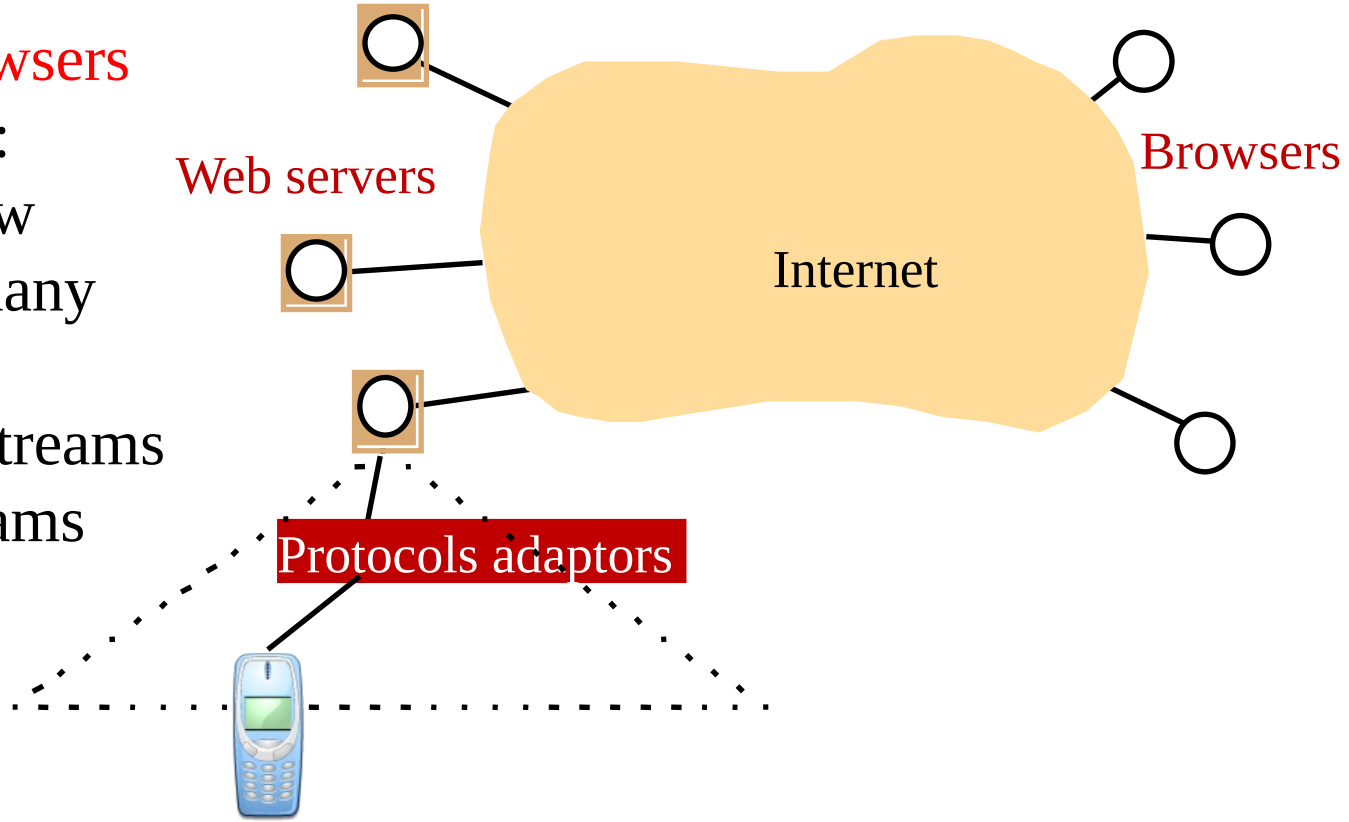
7. The Internet and WWW

- A Web browser is a user interface to the world's biggest DCS, the Internet.
- A Web page includes links to other Web pages. These links are specified as URLs.
- An URL is the name of a protocol (ftp, http, etc.), the name of a site and the name of a file.
- To follow a link to a remote Web page, Web browser talks to the local DNS server to resolve the symbolic site name into an IP address, talks to the http daemon running on that web site and requests the delivery of the Web page addressed by the URL.
- To obtain a file from a remote ftp site, Web browser resolves the site name with the local DNS server

7. The Internet and WWW

Through Web browsers (clients) users can:

- retrieve and view documents of many types
- listen to audio streams
- view video streams
- interact with an unlimited set of services



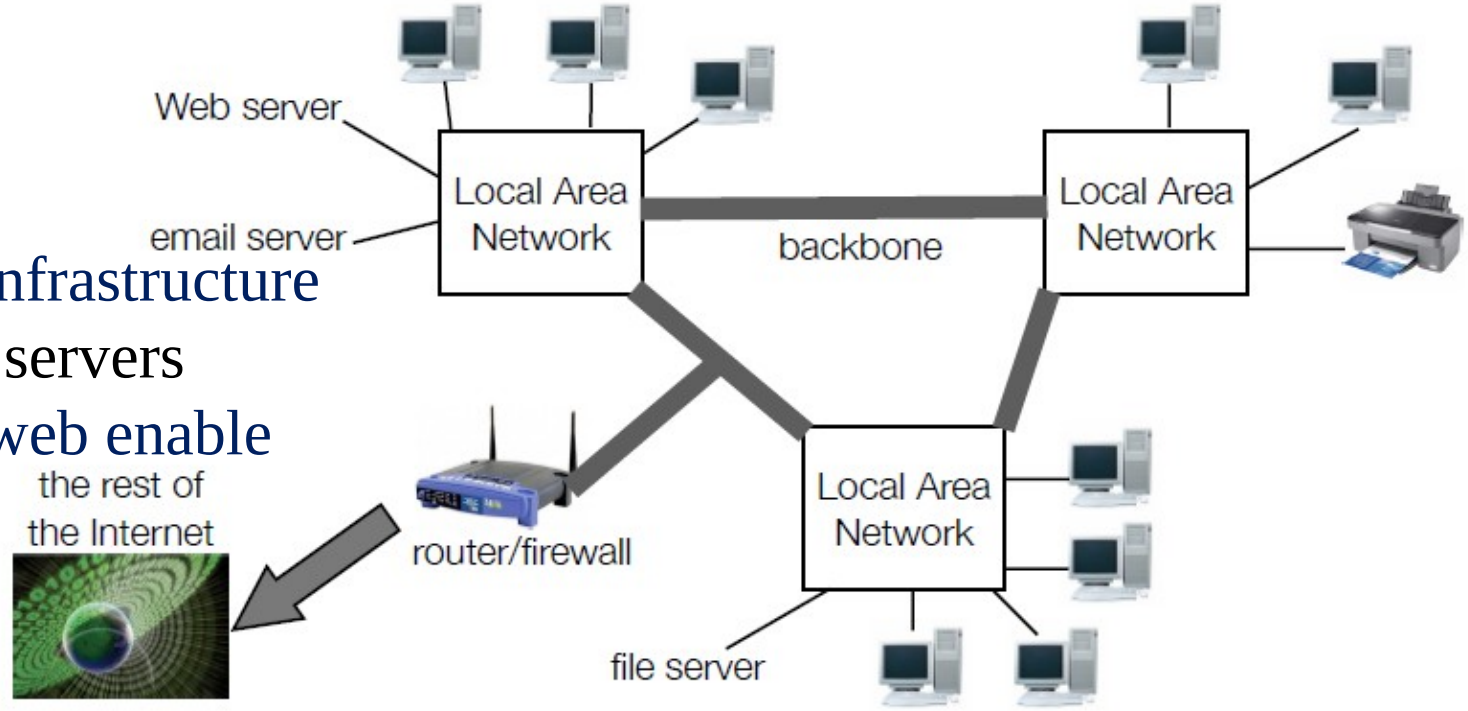
8. The LAN Intranet

Intranet:

1.Network= infrastructure

2.Hardware= servers

3.Software= web enable



A portion of Internet that is separately administered by a company and supports internal sharing of resources (file/storage systems and printers)

8. Local Area Network/Intranet

- One or several print servers control the access to shared printers.
- Clients have the server printing jobs for them.
- Routers provides a gateway to the WAN.
- Users are not aware which machine provides which service.
- Name servers provide the information regarding machine (by IP address), user names (by user ids), and group name (by group ids).
- LANs (Intranets) can be connected together to form Extranets.
- Firewall protect the Internet/Intranet by preventing unauthorized messages leaving or entering the Internet/Intranet





9. Distributed Database Systems

A distributed database is a collection of databases which are distributed over different computers of a computer network.

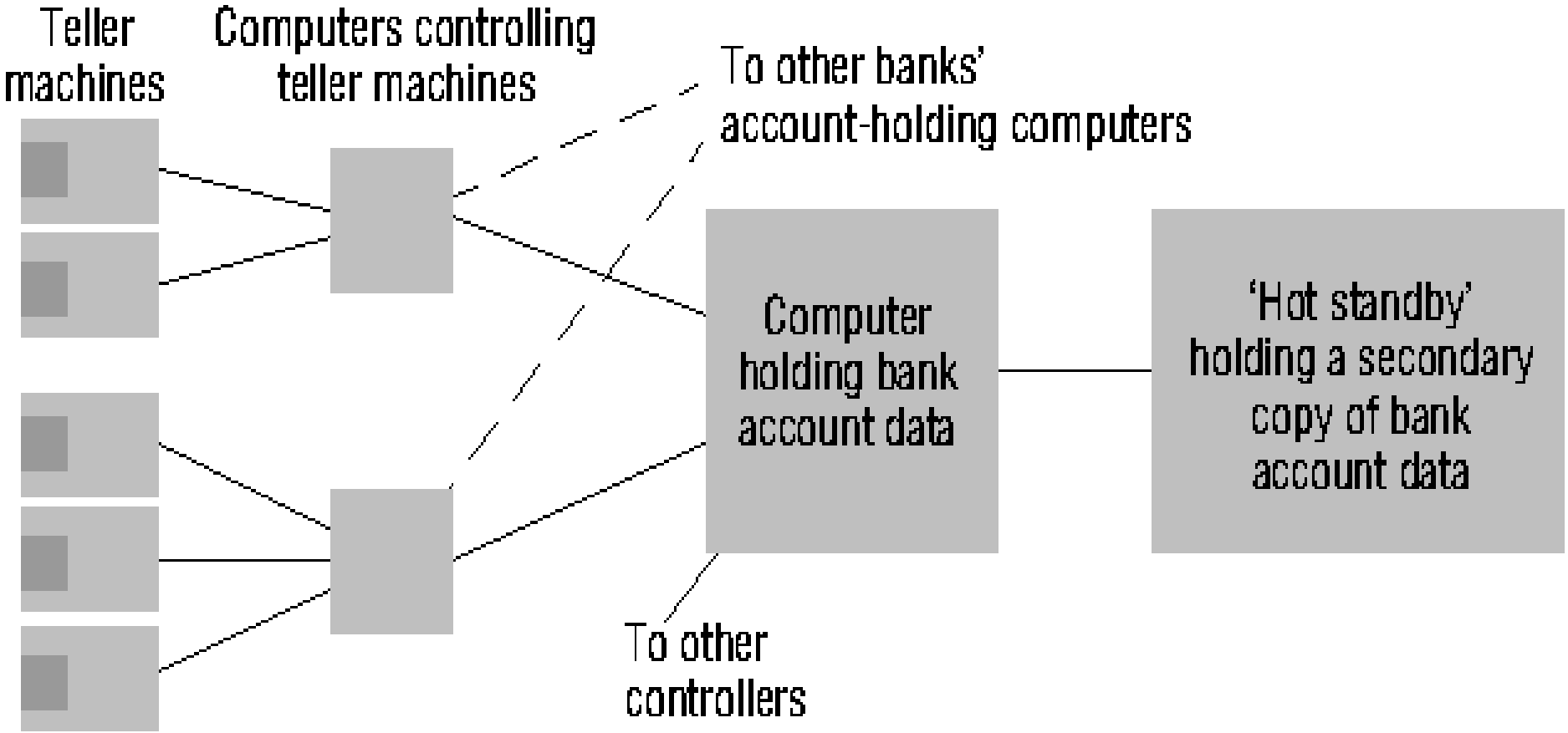
- Each site has autonomous processing capability and can perform local applications
- Each site participate in the execution of at least one global application which requires accessing data at several sites



9. Distributed Database Systems

- Different client applications want to access and update shared data in a database.
- Client applications might be banking systems, real-estate agencies, airline-ticket reservation systems accessing data like balances of bank accounts, details of property that are for sale or to let, or airfares and aircraft reservation data.
- The database is physically distributed over several processors to take advantage of local data accesses for increased performance of client applications.
- Data may be replicated to reduce the impact of failures of a processor and/or the network.
- Each processor runs a database monitor that implements the mapping between the database seen by clients and the physical database stored on the different processors.
- Database monitors have to cooperate with each other to implement client accesses to remote data, updates of replicated data and concurrency control.
- The physical distribution of data is therefore transparent to clients.

10. Automatic Teller Machine Network



10. ATM Network

- An ATM network enables bank customers to withdraw cash from their bank account.
- Banks and building societies maintain large networks of ATM.
- Customer have high security, privacy and reliability requirements.
- Customers may want to withdraw cash from their account through a ´foreign´ ATM.
- A front-end computer controls one or several ATMs. It transfers withdrawal requests to the computer of the account holder's bank, awaits the bank granting the request, and therefore has to be interoperable with heterogeneous computer systems.
- Each bank has fault-tolerant systems to quickly recover from failures of their account holding computers. An example is the ´Hot standby´ computer which maintains a copy of the account database and can replace the main computer within seconds.

11. Mobile Computing

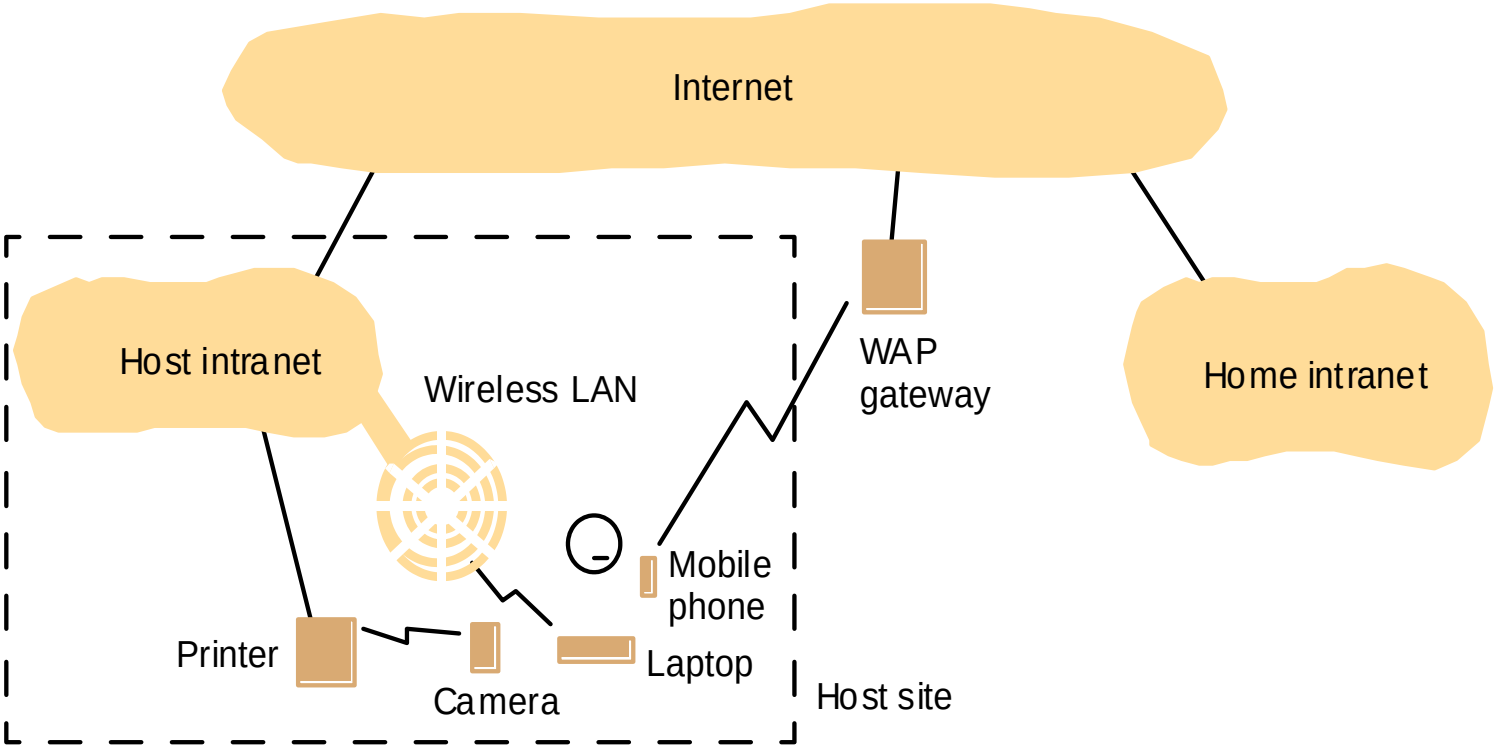
Mobile devices extend the access of Internet and DCS architecture from wire-line connections to wireless connections.



11. Mobile Computing Resource Sharing

- **Mobile computing**: is the technology of using many small, cheap computational devices that are present in users' physical environments, including the home, office and even natural settings.
- Their computational behavior will be transparently and intimately tied up with their physical function.
- The presence of computers everywhere becomes useful when they can communicate with one another.
- E.G. users can control their washing machine or their entertainment system from their phone or a 'universal remote control' device in the home. The washing machine notify the user via an SMS when the washing is done.

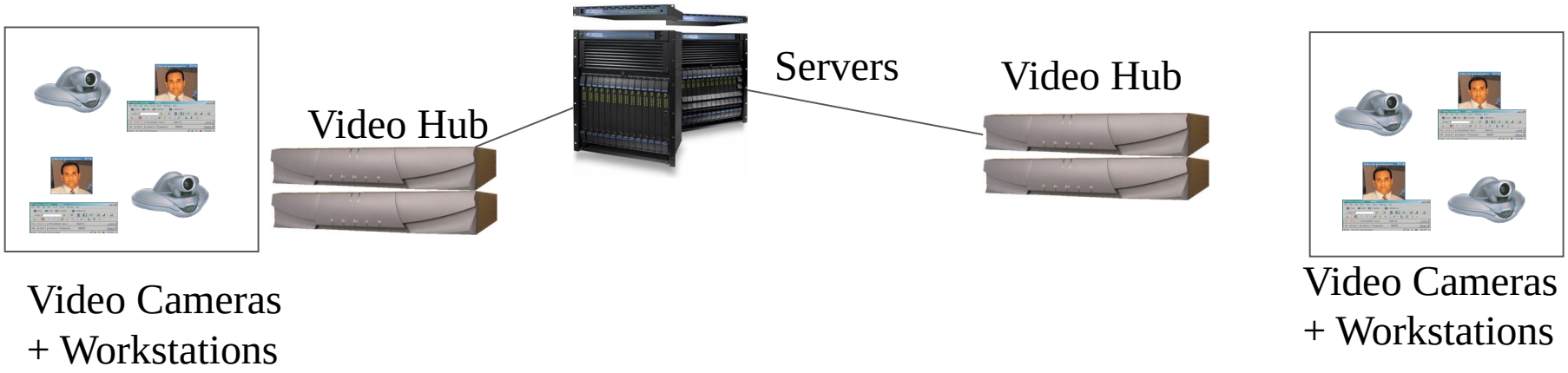
11. Mobile Computing



Portable and handheld devices in a distributed system

12. Video-on-Demand/Teleconferencing Systems

- Client components are used to display videos
- Clients are hosted in the customers workstations
- Servers components load and transmit videos
- Multiple servers are needed in order to enlarge video conferencing

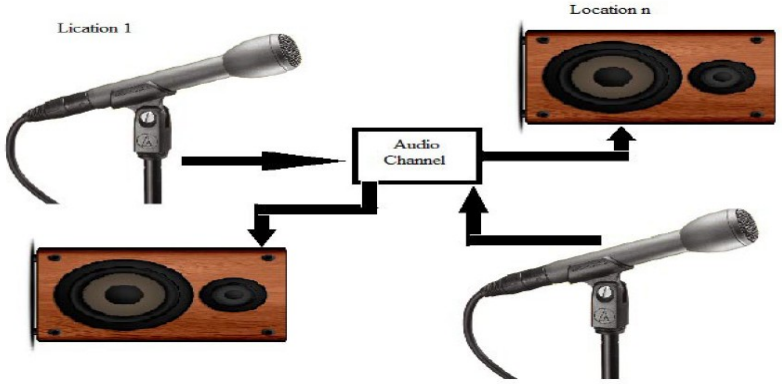


12. Video-on-Demand/Teleconferencing Systems

- It is consists of tools and techniques of both computer and non-computer-based.
- In this techniques that allow a group of people, separated by time and distance, to exchange ideas using audio, video and other teleconferencing media.
- The main feature of this system is that they reduce operating costs and increase productivity because decision makers do not have travel to attend face to face meeting.
- Audio conferencing rules:
  - The person who organizes the conference should serve as a moderator, making certain that all participants have on opportunity to speak and that the objectives of the conference are archived.
  - A copy of the conference agenda should be made available to the participants in advance.

12. Video-on-Demand/Teleconferencing Systems

- Videoconferencing is another type of teleconferencing systems that uses telephones, TV monitors, computers, and networks to link geographically separated decision makers to hear and see each other.
- A computer digitizes sound and video images, then converts them to analog signals and transmits them over the telephone lines to the receiver's computers, which reconverts the analog signals to digital signals.
- These are then translated in to audio and video messages and presented on the TV monitor and sound system.



13. A multiprocessor traffic control system

